Expected Inner Work of Inexact Proximal Point Methods

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Abstract

We consider a monotone inclusion problem in a Hilbert space. Under mild assumptions, it is known that the proximal point method converges to a zero of the operator, even with inexact computations of the resolvent. We extend this analysis to get an explicit $O(1/k)$ nonasymptotic convergence rate of the squared fixed point residual and an asymptotic $o(1/k)$ convergence rate, as well as a linear convergence rate in the presence of an error bound. Assuming access to a linearly convergent iterative procedure to compute the resolvent, we show that the method requires $O(\varepsilon^{-2})$ inner steps to find a point with $\mathbf{dist}(0, Au) \leq \varepsilon$ for a general monotone inclusion problem. We validate our theoretical findings on numerical experiments and show that many popular algorithms are special cases of our framework and benefit from inexact computations.

1 Introduction

In this paper we consider the problem of a maximal monotone inclusion on a real Hilbert space $\mathcal{H}$:

$$\text{find } x \text{ such that } 0 \in Ax, \tag{1}$$

where $A : \mathcal{H} \rightrightarrows \mathcal{H}$ is a maximal monotone operator. Any convex optimization problem can be viewed as a monotone inclusion problem (1) by examining the first-order optimality condition. More broadly, convex-concave minimax problems admit a monotone inclusion formulation. These problems arise in a variety of fields, such as image processing, communications, PDEs, fluid dynamics, and finance [Glowinski et al., 2017].

A popular method for solving the monotone inclusion (1) is the proximal point method (PPM): for a stepsize γ, each iteration solves the backward inclusion $x_k \in (I + \gamma A)x_{k+1}$. However, solving the inclusion subproblem in these algorithms can be difficult and makes each iteration expensive. Rockafellar [1976a] showed that if the sum of all approximation errors is bounded, then PPM still converges asymptotically to a solution of the problem (1). This type of criterion is known as an *absolute* error condition, since it controls the absolute error between the estimated and the true subproblem solution. However, there are several practical issues with such a condition. First, checking the error condition requires knowledge of the true update, which is generally not computable. Second, the choice of such a sequence adds an infinite number of parameters for the user to tune, making practical implementations difficult. Moreover, the convergence analysis does not provide meaningful guidance on the design of such a sequence, nor does it provide guidance on how the degree of error affects the convergence of the algorithm and the complexity of intermediate computation.

To obviate this difficulty, we would like the error condition to have two properties. First, it should be easy to compute, and not require knowledge of the true update. Second, it should not depend on an external sequence of errors, but rather be a function of the current state of the algorithm in some way: a *relative* error condition.

Relaxing the inclusion to allow for error in solving the subproblem makes these algorithms much more efficient in practice. However, the inexactness in solving the subproblem reduces the effectiveness of each iteration of PPM. Thus, there is a tradeoff between exactness and convergence speed (as measured by outer iterations). It is not clear how to tune the parameters of the outer algorithm and the subproblem procedure to balance this tradeoff.

1.1 Contribution

We introduce a unified framework to analyze the per-inner iteration work of inexact PPM and provide guidelines on selecting algorithm parameters to balance the inner and outer iteration complexity. Our setting is general enough to include many popular classes of operating splitting algorithms, such as DRS, DYS, and others. We also allow for any method of solving the inner subproblem, as long as it has linear convergence in expectation. We provide an explicit bound on the total inner iterations required to find a point with $\mathbf{dist}(0, Au) \leq \delta$ in both the deterministic and stochastic cases. Additionally, we provide high-probability bounds in the randomized case for the total complexity. By incorporating warm-starts from previous iterations into the current iteration, we improve the naive worst-case bound of $O(\delta^{-2}\log(1/\delta))$ bound for total inner iteration complexity to $O(\delta^{-2})$ for a general monotone inclusion problem (1).

Our contributions are as follows:

1. We construct a general framework for analyzing degenerate preconditioned proximal point methods and provide bounds on the error at stopping (Proposition 2.1). We improve the analysis of Alves et al. [2026] to get an explicit convergence rate as well as linear convergence in the presence of an error bound (Theorems 3.1 and 3.2).

2. We analyze the total inner work done over the course of the algorithm and show that the method requires $O(\varepsilon^{-2})$ inner steps to find a point with $\mathbf{dist}(0, Au) \leq \varepsilon$ for a general monotone inclusion problem (Theorem 4.2). We also provide high-probability bounds on the total inner work required for such an algorithm (Theorem 4.3).

3. We analyze our results on some model choices of algorithms, such as PPM and ADMM, and infer some principled guidelines on selecting algorithm parameters (Propositions 5.1 to 5.3).

4. **TODO:** update when done We compare our theoretical results against empirical studies and find that the theorems are often conservative. In fact, most instances benefited from decreased accuracy (Section 6).

1.2 Prior Work

Operator Splitting Methods. Martinet [1970] originally proposed PPM for solving variational inequalities, and Rockafellar [1976a,b] greatly advanced theoretical understanding of the method, as well as its relation to other optimization algorithms. Eckstein and Bertsekas [1992] showed that DRS is an instance of PPM applied to a certain splitting operator, showing the connection between PPM and operator splitting algorithms. Operator splitting finds many applications in convex optimization, with notable success stories such as the iterative shrinkage-thresholding algorithm (ISTA), its accelerated variant (FISTA), and the primal-dual hybrid gradient method for linear programming (PDLP) and its variants [Figueiredo and Nowak, 2003; Beck and Teboulle, 2009; Applegate et al., 2021]. Liao et al. [2026] show that under sufficient growth conditions, PPM can even be used for nonconvex optimization as well.

Inexact PPM. One of the challenges in practical implementations of PPM is to solve the backward inclusion subproblem. For example, in the convex optimization setting, this takes the form of solving a strongly convex subproblem. Although first-order methods have theoretically strong guarantees on these problems, solving the problem exactly, or even to high precision, is undesirable. Rockafellar [1976a] showed that it suffices to compute a sequence of iterates whose errors form a summable sequence. Eckstein and Yao [2018] provide an affirmative answer to the question of designing a relative error criterion for Douglas–Rachford and the alternating direction method of multipliers (ADMM), and their result was extended by Svaiter [2019] and Barré et al. [2023].

More recently, Bredies et al. [2022] showed that many existing operator splitting methods, such as Chambolle–Pock, Douglas–Rachford, Peaceman–Rachford, and Davis–Yin, are instances of *degenerate preconditioned* proximal point methods. That is, we precondition the monotone inclusion problem (1) by replacing the scalar stepsize γ with a matrix $M \succeq 0$ and evaluating $(I + M^{-1}A)^{-1} = (M + A)^{-1}M$. Under mild assumptions, even when M is not invertible, the operator $(M+A)^{-1}M$ still exists and can be computed. These ideas can be generalized to relatively inexact resolvent evaluations with a broader class of preconditioners, such as in Alves et al. [2026].

Integrated Complexity Analysis Analyzing the total complexity of the method, which would include the overall number of inner iterations required to solve the inclusion problem, is more delicate. Machart et al. [2012] consider the complexity of inexact proximal gradient methods, but they only consider the case of absolute errors, with the standard summability conditions. Moreover, they only consider a specific instance of PPM, whereas our setting is general enough to cover many more operator splitting methods. Lin et al. [2018] consider a model of global complexity, factoring in inner and outer complexity, but their error criterion is not generally computable, requiring knowledge of the exact solution to the subproblem or access to the dual function. Alves and Geremia [2019, Theorems 4.3 and 4.4] provide analysis in a similar setting but only show asymptotic guarantees of inner complexity. Moreover, they do not consider the effect of warm-starting, resulting in an overall complexity of $O(\delta^{-2}\log(1/\delta))$, compared to our $O(\delta-2)$ result. Li and Qu [2021] consider similar types of inner procedures, but they focus on one particular instance of the augmented Lagrangian method and show worse total complexity guarantees.

2 The Degenerate Preconditioned Proximal Point Method

We make the following assumption to ensure that all the operations we take in this paper are well-defined.

Assumption 2.1. *We assume that A is a maximal monotone operator $A : \mathcal{H} \rightrightarrows \mathcal{H}$ on the Hilbert space $\mathcal{H}$ with at least one zero: $\mathbf{zer}\, A \neq \varnothing$. We assume that the map $C : \mathcal{D} \to \mathcal{H}$ is a bounded linear map between Hilbert spaces, that C^* is surjective, and that $(CC^* + \gamma A)^{-1}CC^*$ is defined everywhere.*

Definition 2.1. Define the operator B by its graph as

$$\mathbf{gra}\, B = \{(C^*u, z) \mid Cz \in Au\}.$$

Since C^* is surjective (Assumption 2.1), the resolvent $J := (I + \gamma B)^{-1}$ is defined on the entire space $\mathcal{D}$ (Lemma 2.1). Note that B itself need not have full domain. In particular, $\mathbf{dom}\, B \subseteq C^*\, \mathbf{dom}\, A$, with strict inclusion a possibility. One way of viewing this definition is that B is a projection of A onto a lower-dimensional space. Proposition 5.3 shows that this perspective enables us to view the iterates of ADMM as the projections of a higher-dimensional PPM.

Lemma 2.1. *For B defined as in Definition 2.1, under Assumption 2.1, the following hold:*

1. $\mathbf{zer}\, B = C^* \mathbf{zer}\, A$.
2. *For $x \in \mathcal{D}$, $w = (I+\gamma B)^{-1}x$ if and only if there exist $v \in \mathcal{H}$ and $z \in \mathcal{D}$ such that $x = C^*v+\gamma z$, $Cz \in Av$, and $w = C^*v$. Moreover, such v and z exist for every $x \in \mathcal{D}$.*
3. *B is maximal monotone.*

Proof. The first claim is immediate. The second follows from Assumption 2.1 and manipulating the definition of B:

$$\begin{aligned}(I+\gamma B)^{-1}x = w &\iff x \in (I+\gamma B)w \\ &\iff x \in (I+\gamma B)w,\ w = C^*v \text{ for some } v \text{ with } C^*v \in \mathbf{dom}\, B \\ &\iff x = C^*v + \gamma z,\ Cz \in Av,\ w = C^*v.\end{aligned}$$

For existence, fix $x \in \mathcal{D}$ and, since C^* is surjective, we can write $x = C^*\tilde{x}$. By Assumption 2.1 there exists v with $CC^*\tilde{x} \in (CC^* + \gamma A)v$, i.e., $\gamma^{-1}CC^*(\tilde{x} - v) \in Av$. Setting $z = \gamma^{-1}C^*(\tilde{x} - v)$ gives $Cz \in Av$ and $C^*v + \gamma z = C^*\tilde{x} = x$.

The third item follows from the fact that

$$\langle C^*u_1 - C^*u_2, z_1 - z_2\rangle = \langle u_1 - u_2, Cz_1 - Cz_2\rangle \geq 0,$$

since $Cz_i \in Au_i$ and A is monotone. Maximality follows from Assumption 2.1: by item 2, $\mathcal{R}(I+\gamma B) = \mathcal{D}$, so B is maximal by Minty's theorem [Bauschke and Combettes, 2017, Theorem 21.1]. □

In the case where A is not necessarily monotone, it may still hold that B is maximal monotone. We make this assumption explicit.

Assumption 2.2. *We assume that B defined in Definition 2.1 is a maximal monotone operator with at least one zero: $\mathbf{zer}\, B \neq \varnothing$. We also assume that the map $C : \mathcal{D} \to \mathcal{H}$ is a bounded linear map between Hilbert spaces with C^* surjective.*

Note that Assumption 2.1 is strictly stronger than Assumption 2.2. Proposition 5.4 shows an example where Assumption 2.2 holds but Assumption 2.1 fails.

Using these assumptions, we can now introduce Algorithm 1, whose convergence properties we will analyze in this section. The following proposition shows that at stopping, the returned iterates provide a certificate of distance to optimality.

Proposition 2.1. *Every output from the $\|z\| \leq \delta$ terminal test of Algorithm 1 satisfies*

$$Cz \in Au, \qquad \mathbf{dist}(0, Au) \leq \|Cz\| \leq \|C\|\delta.$$

Moreover, after stopping we also have

$$\|C^*u - JC^*u\| \leq \gamma\delta.$$

That is, the fixed point residual also satisfies the bound at halting. Note that the returned iterate may not satisfy the relative error criterion.

Algorithm 1 Inexact Proximal Point Method

1: **Initialize:** $w_0 \in \mathcal{D}$, inexactness $\sigma \in [0, 1)$, stepsize $\gamma > 0$, stride $s \in \mathbb{N}$, and stopping tolerance δ
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Produce a pair $(u_k, z_k) \in \mathcal{H} \times \mathcal{D}$ with

$$Cz_k \in Au_k, \qquad \gamma z_k + C^* u_k \approx w_k.$$

4: **if** $\|z_k\| \leq \delta$ **then**
5: Stop and return (z_k, u_k, w_k).
6: **else if** $\|\gamma z_k + C^* u_k - w_k\| \leq \sigma \|C^* u_k - w_k\|$ **then**
7: Set $w_{k+1} = w_k - \gamma z_k$.
8: **else**
9: Improve the pair $(u_k, z_k) \in \mathcal{H} \times \mathcal{D}$ with $Cz_k \in Au_k$ via the iterative procedure s times and go to line 4.
10: **end if**
11: **end for**

Proof. The first claim is immediate from the definition of Algorithm 1. For the second, let $q = C^*u - JC^*u = (I - J)C^*u$. Then,

$$\left(C^*u - q, \frac{q}{\gamma}\right) = \left(JC^*u, \frac{q}{\gamma}\right) = \left(JC^*u, \frac{(I-J)C^*u}{\gamma}\right).$$

Using Assumption 2.2, there exists x such that $C^*x = JC^*u$. Then,

$$\begin{aligned} C^*x = (I + \gamma B)^{-1} C^*u &\iff C^*u \in C^*x + \gamma B C^*x \\ &\iff C^*u = C^*x + \gamma\xi, \quad C\xi \in Ax \\ &\iff \frac{C^*(u-x)}{\gamma} = \xi, \quad C\xi \in Ax. \end{aligned}$$

Thus, we can write

$$\begin{aligned} \left(JC^*u, \frac{(I-J)C^*u}{\gamma}\right) &= (C^*x, \gamma^{-1}C^*(u-x)), \quad \frac{C^*(u-x)}{\gamma} = \xi, C\xi \in Ax \\ &= (C^*x, \xi), \quad \frac{C^*(u-x)}{\gamma} = \xi, C\xi \in Ax, \end{aligned}$$

so $\left(C^*u - q, \frac{q}{\gamma}\right) \in \mathbf{gra}\, B$. Since $(C^*u, z) \in \mathbf{gra}\, B$ and B is monotone,

$$0 \leq \left\langle (C^*u - q) - C^*u, \frac{q}{\gamma} - z \right\rangle = \left\langle q, z - \frac{q}{\gamma} \right\rangle.$$

Thus, $\langle q, \gamma z - q \rangle \geq 0$ and hence

$$\|\gamma z\|^2 = \|\gamma z - q\|^2 + 2\langle \gamma z - q, q \rangle + \|q\|^2 \geq \|q\|^2.$$

Applying the stopping rule $\|z_k\| \leq \delta$ from Algorithm 1 (line 4) completes the proof. □

3 Outer Convergence Guarantees

In this section, we present a quantitative estimate on the convergence of Algorithm 1. The inexactness affects convergence only through a multiplicative factor, which allows the rest of the analysis to follow. The next lemma is needed to show the convergence of Algorithm 1.

Lemma 3.1. *The resolvent J and its fixed-point residual $R := I - J$ are firmly nonexpansive. If $(y, z) \in \mathbf{gra}\, B$ is a candidate update pair (as in line 3 of Algorithm 1 with $y = C^*u$) at w, then*

$$y = J(\gamma z + y), \tag{2}$$
$$\gamma z = R(\gamma z + y), \tag{3}$$
$$\|y - J(w)\|^2 + \|\gamma z - R(w)\|^2 \le \|\gamma z + y - w\|^2. \tag{4}$$

If the pair (y, z) passes the relative test, then we also have

$$(1-\sigma)\|y - w\| \le \|R(w)\| \le (1+\sigma)\|y - w\|, \tag{5}$$
$$(1-\sigma)\|y - w\| \le \|\gamma z\| \le (1+\sigma)\|y - w\|, \tag{6}$$
$$\|\gamma z + y - w\| \le \frac{\sigma}{1-\sigma}\|R(w)\|, \tag{7}$$
$$\|\gamma z\| \le \frac{\|R(w)\|}{1-\sigma}. \tag{8}$$

Moreover, $\|R(w)\| \le \mathbf{dist}\,(w, \mathbf{zer}\, B)$.

Proof. For the first equation, note that $y = C^*u$ and thus

$$y = J(\gamma z + y) \iff \gamma z + y \in y + \gamma B y \iff z \in BC^*u,$$

which is the definition of B. Next,

$$R(\gamma z + y) = \gamma z + y - J(\gamma z + y) = \gamma z + y - y = \gamma z.$$

Firm nonexpansiveness follows from Assumption 2.2 and the firm nonexpansiveness of resolvents of maximal monotone operators [Bauschke and Combettes, 2017, Corollary 23.11].

Next, by the monotonicity of B,

$$\langle y_1 - y_2, z_1 - z_2 \rangle \ge 0,$$

so $\langle y_1 - y_2, \gamma z_1 - \gamma z_2 \rangle \ge 0$ as well. For an exact pair, $w = y + \gamma z$ with $z \in By$,

$$\begin{aligned} \|w_1 - w_2\|^2 &= \|\gamma z_1 - \gamma z_2\|^2 + \|y_1 - y_2\|^2 + 2\langle y_1 - y_2, \gamma z_1 - \gamma z_2 \rangle \\ &\ge \|\gamma z_1 - \gamma z_2\|^2 + \|y_1 - y_2\|^2. \end{aligned} \tag{9}$$

Note that since $w = y + \gamma z$ with $z \in By$, we have $y = J(w)$ and $\gamma z = R(w)$. That is, (9) states $\|w_1 - w_2\|^2 \ge \|R(w_1) - R(w_2)\|^2 + \|J(w_1) - J(w_2)\|^2$ for all w_1 and w_2.

Let $e = \gamma z + y - w$, the graph residual. Note that $w + e = \gamma z + y$. Since $y = J(\gamma z + y) = J(w + e)$,

$$\gamma z = w + e - y = w + e - J(w + e) = (I - J)(w + e) = R(w + e).$$

Thus, using the resolvent identities $y = J(\gamma z + y)$ and $\gamma z = R(\gamma z + y)$ (2) and (3) and the component bound $\|w_1 - w_2\|^2 \geq \|\gamma z_1 - \gamma z_2\|^2 + \|y_1 - y_2\|^2$ (9) with $w_1 = w + e$ and $w_2 = w$,

$$\begin{aligned}\|e\|^2 = \|(w+e) - w\|^2 &\geq \|R(w+e) - R(w)\|^2 + \|J(w+e) - J(w)\|^2 \\ &= \|R(\gamma z + y) - R(w)\|^2 + \|J(\gamma z + y) - J(w)\|^2 \\ &= \|\gamma z - R(w)\|^2 + \|y - J(w)\|^2.\end{aligned}$$

Now suppose the relative test $\|e\| \leq \sigma\|y - w\|$ holds. By the reverse triangle inequality,

$$|\|y - w\| - \|R(w)\|| = |\|y - w\| - \|w - J(w)\|| \leq \|y - J(w)\| \leq \|e\| \leq \sigma\|y - w\|.$$

This is the residual bound (5). Next, $\gamma z = e - (y - w)$, so the triangle inequality gives

$$(1 - \sigma)\|y - w\| \leq \|y - w\| - \|e\| \leq \|\gamma z\| \leq \|y - w\| + \|e\| \leq (1 + \sigma)\|y - w\|,$$

which is the update bound (6). Combining $\|e\| \leq \sigma\|y - w\|$ with the left inequality of (5) gives the error bound (7). Finally, $\|\gamma z - R(w)\| \leq \|e\|$ and $\|e\| \leq \frac{\sigma}{1-\sigma}\|R(w)\|$ by the component bound (4) and the error bound (7), so

$$\|\gamma z\| \leq \|R(w)\| + \|\gamma z - R(w)\| \leq \|R(w)\| + \|e\| \leq \left(1 + \frac{\sigma}{1 - \sigma}\right)\|R(w)\| = \frac{\|R(w)\|}{1 - \sigma},$$

which is the bound (8).

Lastly, if $w^\star \in \mathbf{zer}\, B$, then $Jw^\star = w^\star$. Thus, $R(w^\star) = 0$, so by the nonexpansiveness of R,

$$\|R(w)\| = \|R(w) - R(w^\star)\| \leq \|w - w^\star\|.$$

Taking infimums shows the desired result. □

Similar results to Proposition 2.1 and Lemma 3.1 can be found in Rockafellar [1976b, Propositions 1, 2, and 3], Bredies et al. [2022, Lemma 2.6], and Alves et al. [2026, Proposition 2.3] (among others), but we present additional identities and estimates here for our purposes.

Lemma 3.2. *For an accepted pair with $w^+ = w - \gamma z$ and any $w^\star \in \mathbf{zer}\, B$,*

$$\|w^+ - w^\star\|^2 \leq \|w - w^\star\|^2 - \|y - w\|^2 + \|\gamma z + y - w\|^2.$$

Under the relative error rule, this yields

$$\begin{aligned}\|w^+ - w^\star\|^2 &\leq \|w - w^\star\|^2 - \frac{\gamma^2(1 - \sigma)}{1 + \sigma}\|z\|^2, \\ \|w^+ - w^\star\|^2 &\leq \|w - w^\star\|^2 - \frac{1 - \sigma}{1 + \sigma}\|R(w)\|^2.\end{aligned}$$

Proof. Recall that we defined $e = \gamma z + y - w$, the graph residual. The claim then follows immediately from expanding $w = \gamma z + y - e$:

$$\begin{aligned}\|w - w^\star\|^2 &= \|w - \gamma z - w^\star\|^2 + 2\langle w - \gamma z - w^\star, \gamma z\rangle + \|\gamma z\|^2 \\ &= \|w - \gamma z - w^\star\|^2 + 2\langle y - e - w^\star, \gamma z\rangle + \|\gamma z\|^2 \\ &= \|w - \gamma z - w^\star\|^2 + 2\langle y - w^\star, \gamma z\rangle - 2\langle e, \gamma z\rangle + \|\gamma z\|^2.\end{aligned}$$

Using the monotonicity of B (with $(w^\star, 0), (y, z) \in \mathbf{gra}\, B$) on the first inner product term and then completing the square yields

$$\begin{aligned}\|w - w^\star\|^2 &\geq \|w - \gamma z - w^\star\|^2 - \|e\|^2 + \|e - \gamma z\|^2 \\ &= \|w - \gamma z - w^\star\|^2 - \|\gamma z + y - w\|^2 + \|w - y\|^2.\end{aligned}$$

Applying the relative error rule, $\|e\| \leq \sigma\|y - w\|$, and Lemma 3.1 yields

$$\begin{aligned}\|w - \gamma z - w^\star\|^2 &\leq \|w - w^\star\|^2 + \|\gamma z + y - w\|^2 - \|w - y\|^2 \\ &\leq \|w - w^\star\|^2 + \sigma^2\|y - w\|^2 - \|w - y\|^2 \\ &= \|w - w^\star\|^2 - (1 - \sigma^2)\|w - y\|^2 \\ &\leq \|w - w^\star\|^2 - \frac{(1 - \sigma^2)}{(1 + \sigma)^2}\|\gamma z\|^2.\end{aligned}$$

The last inequality uses $\|\gamma z\| \leq (1 + \sigma)\|y - w\|$ from the update bound (6) of Lemma 3.1, which yields the first bound. Using $\|R(w)\| \leq (1 + \sigma)\|y - w\|$ from the residual bound (5) of Lemma 3.1 instead gives the second. □

Using Lemma 3.2, we immediately obtain the following finite termination guarantee on Algorithm 1. We remark that this matches well-known results on the $o(k^{-1})$ convergence rate of the squared fixed point residual.

Theorem 3.1. *After K accepted updates,*

$$\sum_{k=0}^{K-1} \|\gamma z_k\|^2 \leq \frac{(1 + \sigma)\,\mathbf{dist}\,(w_0, \mathbf{zer}\, B)^2}{1 - \sigma}.$$

As a result,

$$\min_{k=0,\ldots,K-1} \|C z_k\| \leq \frac{\|C\|\sqrt{1 + \sigma}\,\mathbf{dist}\,(w_0, \mathbf{zer}\, B)}{\gamma\sqrt{(1 - \sigma)K}}.$$

Moreover, if K_δ is the number of accepted updates before returning a point satisfying $\|z_{K_\delta}\| \leq \delta$ (the terminal test of Algorithm 1), then

$$K_\delta \leq \left\lfloor \frac{(1 + \sigma)\,\mathbf{dist}\,(w_0, \mathbf{zer}\, B)^2}{\gamma^2\delta^2(1 - \sigma)} \right\rfloor =: K_\delta^+.$$

Moreover, if $K_\delta > 0$, then $K_\delta < K_\delta^+$.

Proof. The first two claims follow immediately from the telescoping identity of Lemma 3.2 and Proposition 2.1. The last claim follows from the fact that for all iterations before K_δ, the accepted iterate satisfies $\|z_k\| > \delta$. Therefore, applying the first claim with $K = K_\delta$,

$$\frac{(1 + \sigma)\,\mathbf{dist}\,(w_0, \mathbf{zer}\, B)^2}{1 - \sigma} \geq \sum_{k=0}^{K_\delta - 1} \|\gamma z_k\|^2 \geq K_\delta \gamma^2 \delta^2,$$

where the second inequality is strict if $K_\delta \geq 1$ and trivial if $K_\delta = 0$. Recognizing that K_δ is an integer and rearranging yields the result. □

Corollary 3.1.1. *Using Theorem 3.1 and setting $\delta = 0$, we find $\min_{k=0,\ldots,K-1}\|z_k\|^2 = o(K^{-1})$.*

Proof. From the first part of Theorem 3.1, we know that $\sum_{k=0}^{\infty}\|\gamma z_k\|^2 \leq \frac{(1+\sigma)\,\mathbf{dist}\,(w_0,\mathbf{zer}\,B)^2}{1-\sigma}$. Thus,

$$\frac{K}{2}\min_{k=0,\ldots,K-1}\|\gamma z_k\|^2 \leq \sum_{k=\lfloor\frac{K}{2}\rfloor}^{K-1}\|\gamma z_k\|^2 \leq \sum_{k=\lfloor\frac{K}{2}\rfloor}^{\infty}\|\gamma z_k\|^2 \to 0$$

as $K\to\infty$. Since $\gamma>0$, we have the desired asymptotic convergence rate. □

Remark 3.1. By Lemma 3.1, bounds on $\|\gamma z_k\|$ can be converted into bounds on $\|R(w)\|$. For brevity, we omit them here.

In essence, Theorem 3.1 shows how many iterations are required before stopping, and Proposition 2.1 guarantees the accuracy of the iterate at halting. We now show that under an error bound guarantee, we obtain linear convergence of the method, where the contraction factor is attenuated by the inexactness of the method.

Theorem 3.2. *Under an error bound guarantee of the form*

$$\mathbf{dist}(w_k,\mathbf{zer}\,B)\leq\Lambda\|R(w_k)\|$$

for all k, where $\Lambda\geq 1$, we find that

$$\mathbf{dist}(w_k,\mathbf{zer}\,B)^2\leq\left(1-\frac{1-\sigma}{\Lambda^2(1+\sigma)}\right)^k\mathbf{dist}(w_0,\mathbf{zer}\,B)^2.$$

Moreover, the stopping time $K_\delta=\min\{k\mid\|z_k\|\leq\delta\}$ obeys

$$K_\delta\leq\frac{\log\left[\mathbf{dist}(w_0,\mathbf{zer}\,B)^2\right]-\log\left[(1-\sigma)^2\gamma^2\delta^2\right]}{\log\frac{1}{\omega}},$$

where $\omega=1-\frac{1-\sigma}{\Lambda^2(1+\sigma)}$.

Proof. From Lemma 3.2,

$$\begin{aligned}\mathbf{dist}(w^+,\mathbf{zer}\,B)^2&\leq\mathbf{dist}(w,\mathbf{zer}\,B)^2-\frac{1-\sigma}{1+\sigma}\|R(w)\|^2\\&\leq\left(1-\frac{1-\sigma}{\Lambda^2(1+\sigma)}\right)\mathbf{dist}(w,\mathbf{zer}\,B)^2.\end{aligned}$$

Unraveling the recursion completes the proof of the first inequality.

Next, by the bound (8) of Lemma 3.1, $\|z_k\|\leq\frac{\mathbf{dist}(w_k,\mathbf{zer}\,B)}{\gamma(1-\sigma)}$ for updates passing the relative test. Thus, a sufficient condition for stopping is $\mathbf{dist}(w_k,\mathbf{zer}\,B)\leq(1-\sigma)\gamma\delta$, which happens after at most

$$\frac{\log\left[\mathbf{dist}(w_0,\mathbf{zer}\,B)^2\right]-\log\left[(1-\sigma)^2\gamma^2\delta^2\right]}{\log\frac{1}{\omega}}$$

accepted updates, with $\omega=1-\frac{1-\sigma}{\Lambda^2(1+\sigma)}$. □

Remark 3.2. The convergence results of Theorems 3.1 and 3.2 can be modified to cover the case of deterministic time-varying stepsizes as well. This case is a simple application of Lemma 3.2, and we omit it here.

4 Inner Complexity Properties

The analysis of Section 3 ignores the fact that solving the proximal subproblem is usually not trivial. For example, the proximal operation on the quadratic function $f(x) = \frac{1}{2}x^\top Qx + d^\top x$ requires solving the system $(\gamma Q + I)x^+ = x - \gamma d$. This can be nontrivial in general, and it would be ideal to obtain guarantees for the complexity of the algorithm in terms of the number of inner iterations, as well as to connect this to the general structure of the problem and parameters.

We begin with some definitions to establish the setting of the problem.

Definition 4.1. Let $\mathbf{P}_k(\mathcal{E}) := \mathbf{P}(\mathcal{E} \mid \mathcal{F}_k)$, and $\mathbf{E}_k$ similarly. Here $\{\mathcal{F}_k\}$ represents the filtration generated by the randomness of the algorithm up to the start of *outer* iteration k, i.e., all inner and outer history before the kth inner solve begins. In particular w_k, $R(w_k)$, the warm start of the kth inner solve, and the event $\{K_\delta \geq k\}$ are $\mathcal{F}_k$-measurable. We write N_k for the number of inner iterations performed during outer iteration k.

Assumption 4.1. *We assume that the inner procedure referred to in Algorithm 1 has a guarantee of the form*

$$\mathbf{E}_k[\|\gamma z^l + y^l - w_k\|^2] \leq C_0 \alpha^l \mathbf{E}_k[\|\gamma z^0 + y^0 - w_k\|^2],$$

where $0 < \alpha < 1$ and $C_0 \geq 1$ are constants that do not depend on k or l. Here the superscripts represent inner *iterations (those in the loop of lines 4–10 in Algorithm 1).*

This may seem like a strong assumption, since it asks for an inner procedure whose residual contracts linearly.[1] It is, however, mild. When $A = \partial f$ for a closed, proper, and convex function f, the resolvent is the proximal operator of f. Thus, the subproblem $\min_x f(x) + \frac{1}{2\gamma}\|x - w_k\|^2$ is $\frac{1}{\gamma}$-strongly convex, so any linearly convergent method for strongly convex problems qualifies. For smooth f, gradient descent is one such inner procedure, with the constants of Proposition 5.1, and Proposition 5.3 giving the constants for the x-subproblem of ADMM. In the quadratic example above, where $f(x) = \frac{1}{2}x^\top Qx + d^\top x$ and the subproblem is to solve the linear system $(\gamma Q + I)x^+ = x - \gamma d$, the generalized minimal residual method (GMRES) contracts the residual linearly, with the constants used in Section 6.2. A method that does not converge linearly on a strongly convex subproblem would be a poor choice of inner solver.

Theorem 4.1. *Under Assumptions 2.2 and 4.1, if $\sigma \in (0, 1)$, we have the following bounds on the conditional expectation of the inner work of Algorithm 1 at outer iteration k, for every integer $l \geq 0$:*

$$\mathbf{P}_k(N_k > sl) \leq \min\{1, \beta_k \alpha^{sl}\},$$
$$\mathbf{E}_k[N_k] \leq \frac{(\log \beta_k)_+}{\log(1/\alpha)} + \frac{s}{1 - \alpha^s},$$

where

$$\beta_k = \frac{(1+\sigma)^2 C_0 \mathbf{E}_k[\|\gamma z^0 + y^0 - w_k\|^2]}{\sigma^2 \max\{\|R(w_k)\|^2, \gamma^2\delta^2\}}.$$

Proof. On the event $\{N_k > sl\}$, both tests in Algorithm 1 fail at the check after sl inner iterations. Write $y = C^* u^{sl}$ and $e = \gamma z^{sl} + y - w_k$. Thus, $\|\gamma z^{sl} + C^* u^{sl} - w_k\| > \sigma\|C^* u^{sl} - w_k\|$. Note that Lemma 3.1 gives $\|y - J(w_k)\| \leq \|e\|$, so

$$\|R(w_k)\| \leq \|y - w_k\| + \|y - J(w_k)\| < \frac{\|e\|}{\sigma} + \|e\| = \frac{1+\sigma}{\sigma}\|e\|.$$

[1] The term "linear convergence" is often referred to as "geometric convergence" outside of optimization contexts.

Similarly, since $\gamma z^{sl} = e - (y - w_k)$ and the terminal test fails ($\|z^{sl}\| > \delta$),

$$\gamma\delta < \|\gamma z^{sl}\| \le \|e\| + \|y - w_k\| < \frac{1+\sigma}{\sigma}\|e\|.$$

Thus, by the conditional Markov's inequality,

$$\begin{aligned}
\mathbf{P}_k(N_k > sl) &\le \mathbf{P}_k\left(\|\gamma z^{sl} + C^* u^{sl} - w_k\| > \frac{\sigma}{1+\sigma}\max\{\|R(w_k)\|, \gamma\delta\}\right) \\
&\le \frac{\mathbf{E}_k[\|\gamma z^{sl} + C^* u^{sl} - w_k\|^2]}{\left(\frac{\sigma}{1+\sigma}\max\{\|R(w_k)\|, \gamma\delta\}\right)^2} \\
&\le \beta_k \alpha^{sl}.
\end{aligned}$$

The expectation bound holds since N_k takes values $0, s, 2s, \ldots$ and thus

$$\begin{aligned}
\mathbf{E}_k[N_k] = \sum_{l=0}^{\infty} \mathbf{P}_k(N_k > l) &\le s \sum_{l=0}^{\infty} \min\{1, \beta_k \alpha^{sl}\} \\
&= s\left\lceil \frac{(\log \beta_k)_+}{s\log(1/\alpha)} \right\rceil + s\beta_k \sum_{l=\left\lceil \frac{(\log \beta_k)_+}{s\log(1/\alpha)} \right\rceil}^{\infty} \alpha^{sl} \\
&= s\left\lceil \frac{(\log \beta_k)_+}{s\log(1/\alpha)} \right\rceil + s\beta_k \frac{\alpha^{s\cdot\left\lceil \frac{(\log \beta_k)_+}{s\log(1/\alpha)} \right\rceil}}{1-\alpha^s} \\
&\le \frac{(\log \beta_k)_+}{\log(1/\alpha)} + \frac{s}{1-\alpha^s}.
\end{aligned}$$

For the last step, let $l^\star := \frac{(\log \beta_k)_+}{s\log(1/\alpha)}$ and note that $\lceil l^\star \rceil - l^\star \in [0, 1)$. If $\beta_k \ge 1$, then $\beta_k \alpha^{sl^\star} = 1$, so $\beta_k \alpha^{s\lceil l^\star \rceil} = \alpha^{s(\lceil l^\star \rceil - l^\star)}$ and the previous line equals $s\left(l^\star + (\lceil l^\star \rceil - l^\star) + \frac{\alpha^{s(\lceil l^\star \rceil - l^\star)}}{1-\alpha^s}\right)$. The function $x \mapsto x + \frac{\alpha^{sx}}{1-\alpha^s}$ is convex on $[0, 1]$ and equals $\frac{1}{1-\alpha^s}$ at both endpoints, so it is at most $\frac{1}{1-\alpha^s}$ on $[0, 1)$. If $\beta_k < 1$, then $l^\star = 0$ and the sum equals $\frac{\beta_k}{1-\alpha^s} \le \frac{1}{1-\alpha^s}$. In both cases, $\sum_{l=0}^{\infty} \min\{1, \beta_k \alpha^{sl}\} \le l^\star + \frac{1}{1-\alpha^s}$, which is the claim. □

4.1 Warm Start Analysis

The analysis in Theorem 4.1 shows that the inner stopping time N_k has exponentially decaying tails and thus is finite almost surely. However, the structure of the guarantee in Assumption 4.1 implies that the inner method is sensitive to the choice of initialization. In fact, this is one of the motivations for using the PPM: we can solve a regularized subproblem at each iteration and warm-start the inner solver with the previous iterate. The following lemma bounds the drift of the solution due to this choice of warm-starting.

Lemma 4.1. *Suppose that the inner solve in Algorithm 1 is warm-started at the previously accepted solution. Let (u_k, z_k) be the pair accepted at outer iteration k (as in Algorithm 1), $y_k = C^* u_k$, and $e_k = \gamma z_k + y_k - w_k$, so that $w_{k+1} = w_k - \gamma z_k$. Then, we have the following bounds on the drift of the inner system:*

$$\|\gamma z_k + C^* u_k - w_{k+1}\| \le G_\sigma \|R(w_k)\|, \tag{10a}$$

$$\|R(w_{k+1})\| \le H_\sigma \|R(w_k)\|, \tag{10b}$$

where, with $\sigma_1 := \frac{\sqrt{13}-1}{4}$ *and* $\sigma_2 := \frac{\sqrt{7}-1}{2}$,

$$H_\sigma := \frac{1}{\sqrt{1-\sigma^2}}, \qquad G_\sigma := \begin{cases} 1+2\sigma, & 0 < \sigma \le \sigma_1, \\ 2\sqrt{\dfrac{3(1-\sigma^2)}{3-4\sigma^2}}, & \sigma_1 \le \sigma \le \sigma_2, \\ \dfrac{2\sigma-1}{1-\sigma}, & \sigma_2 \le \sigma < 1. \end{cases} \tag{11}$$

The function $\sigma \mapsto G_\sigma$ *is continuous and increasing, and* $G_\sigma \ge H_\sigma$. *Moreover, both constants* H_σ *and* G_σ *in (10) are tight.*

The proof can be found in Appendix B.

Corollary 4.1.1. *Conditionally, for* $k \ge 1$ *on the event* $\{K_\delta \ge k\}$,

$$\beta_k \le \frac{D_\sigma \|R(w_{k-1})\|^2}{\max\{\|R(w_k)\|^2, \gamma^2\delta^2\}},$$

where

$$D_\sigma := \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2}.$$

Combining Theorem 4.1 and Lemma 4.1 yields a bound on the total inner work done to reach a residual of size at most δ.

Theorem 4.2. *Suppose Assumptions 2.2 and 4.1 hold,* $\sigma \in (0,1)$, *and for* $k \ge 1$ *the inner solver is warm-started at the previously accepted pair as in Lemma 4.1. Let* $D = \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2}$ *as in Corollary 4.1.1, and let* β_0 *be the quantity of Theorem 4.1 for the (cold-started) first inner solve. Then*

$$\mathbf{E}\left[\sum_{k=0}^{K_\delta} N_k\right] \le \mathbf{E}[K_\delta]\left(\frac{s}{1-\alpha^s} + \frac{\log D_\sigma}{\log(1/\alpha)}\right) + \frac{1}{\log(1/\alpha)}\left(\frac{s\log(1/\alpha)}{1-\alpha^s} + (\log\beta_0)_+ + \log\frac{\max\left\{\|R(w_0)\|^2, \gamma^2\delta^2\right\}}{\gamma^2\delta^2}\right). \tag{12}$$

As a consequence, $\mathbf{E}\left[\sum_{k=0}^{K_\delta} N_k\right] = O(\delta^{-2})$.

Proof. Let $r_k := \max\{\|R(w_k)\|, \gamma\delta\}$, $c_1 := \frac{s}{1-\alpha^s}$, and

$$T_k := c_1 + \frac{(\log\beta_k)_+}{\log(1/\alpha)}.$$

By Theorem 4.1, $\mathbf{E}_k[N_k] \le T_k$ on $\{K_\delta \ge k\}$, and T_k is $\mathcal{F}_k$-measurable.

Fix $k \ge 1$ on $\{K_\delta \ge k\}$. By Corollary 4.1.1, $\beta_k \le \frac{D r_{k-1}^2}{r_k^2}$. By Lemma 4.1 and the fact that $H_\sigma \ge 1$, $r_k \le H_\sigma r_{k-1}$. Thus,

$$\log D_\sigma + \log\frac{r_{k-1}^2}{r_k^2} \ge \log\frac{D}{H_\sigma^2} \ge 0,$$

since $D = \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2} \geq G_\sigma^2 \geq H_\sigma^2$. Here $C_0 \geq 1$, $\frac{1+\sigma}{\sigma} > 1$, and $G_\sigma \geq H_\sigma$ by Lemma 4.1. Therefore, the maximum with 0 in the $(\log)_+$ term can be dropped:

$$(\log \beta_k)_+ \leq \max\left\{0, \log D_\sigma + \log \frac{r_{k-1}^2}{r_k^2}\right\} = \log D_\sigma + \log \frac{r_{k-1}^2}{r_k^2}.$$

Summing and telescoping, for every $K \leq K_\delta$,

$$\sum_{k=1}^{K} (\log \beta_k)_+ \leq K \log D_\sigma + \log \frac{r_0^2}{r_K^2} \leq K \log D_\sigma + \log \frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}, \tag{13}$$

using $r_K \geq \gamma\delta$ to eliminate the maximum in the denominator. Taking $K = K_\delta$ gives the almost sure bound

$$\sum_{k=0}^{K_\delta} T_k \leq (K_\delta + 1)c_1 + \frac{1}{\log(1/\alpha)}\left((\log \beta_0)_+ + K_\delta \log D_\sigma + \log \frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}\right).$$

By Tonelli's theorem and the tower property,

$$\mathbf{E}\left[\sum_{k=0}^{K_\delta} N_k\right] = \sum_{k=0}^{\infty} \mathbf{E}\left[\mathbf{1}_{\{K_\delta \geq k\}} \mathbf{E}_k[N_k]\right] \leq \sum_{k=0}^{\infty} \mathbf{E}\left[\mathbf{1}_{\{K_\delta \geq k\}} T_k\right] = \mathbf{E}\left[\sum_{k=0}^{K_\delta} T_k\right].$$

Taking expectations in the almost sure bound yields the theorem. All quantities are finite because $K_\delta \leq K_\delta^+$ almost surely (Theorem 3.1).

Lastly, by Theorem 3.1, $K_\delta \leq K_\delta^+ = O(\delta^{-2})$. Thus, $\mathbf{E}\left[\sum_{k=0}^{K_\delta} N_k\right] = O(\delta^{-2})$. □

Remark 4.1. The $\log(\delta^{-1})$ term cannot in general be omitted from a bound in terms of the actual number of outer updates. Proposition A.1 gives a fixed smooth convex instance with $K_\delta = 1$ whose total inner work has exactly the logarithmic leading term in (12). Separately, Proposition A.2 gives a family attaining $\Theta(\delta^{-2})$ total inner work with fixed algorithm parameters and bounded initialization quantities.

The choice of warm start is flexible. The main ingredient in the proof is a bound on the initialization in terms of the previous fixed point residual and a bound on the next fixed point residual in terms of the current. That is, it suffices to find constants G and H such that

$$\begin{aligned} \mathbf{E}_{k+1}[\|\gamma z^0 + C^* u^0 - w_{k+1}\|^2] &\leq G^2 \|R(w_k)\|^2, \quad \text{(warm-start bound)} \\ \|R(w_{k+1})\| &\leq H \|R(w_k)\|, \quad \text{(residual drift bound)} \end{aligned}$$

where $H \geq 1$. The proof of Theorem 4.2 can be repeated with D_σ replaced by $\max\left\{\frac{(1+\sigma)^2 C_0 G^2}{\sigma^2}, H^2\right\}$. The maximum with H^2 is what allows the $\max\{0, \cdot\}$ in $(\log)_+$ to be dropped. Lemma 4.1 gives $G = G_\sigma$ and $H = H_\sigma$ from (10), for which the maximum is D_σ. In fact, the proof of Theorem 4.2 used few properties of the outer algorithm in which it was implemented. The main ingredients of the proof are estimates on the drift of the inner subproblem from iteration to iteration. It is an interesting question to see if it is possible to apply this strategy to other well-known algorithms which assume some kind of (possibly inexact) subproblem oracle.

Theorem 4.1 also yields exponential concentration of the total inner work. The proof uses the exponential supermartingale method described by Howard et al. [2020].

Theorem 4.3 (Concentration of total inner work)**.** *We define the upper bound function $\mathcal{T}_\delta$ of Theorem 4.2 as*

$$\mathcal{T}_\delta(m) := m\left(\frac{s}{1-\alpha^s} + \frac{\log D_\sigma}{\log(1/\alpha)}\right) + \frac{1}{\log(1/\alpha)}\left(\frac{s\log(1/\alpha)}{1-\alpha^s} + (\log\beta_0)_+ + \log\frac{\max\left\{\|R(w_0)\|^2, \gamma^2\delta^2\right\}}{\gamma^2\delta^2}\right),$$

with $(\log 0)_+ := 0$*. Under the assumptions of Theorem 4.2, for every* $\eta \in (0,1)$*,*

$$\mathbf{P}\left(\sum_{k=0}^{K_\delta} N_k > \mathcal{T}_\delta(K_\delta) + \frac{\sqrt{2(K_\delta^+ + 1)\log(1/\eta)} + \log(1/\eta)}{\log(1/\alpha)}\right) \le \eta, \tag{14}$$

$$\mathbf{P}\left(\sum_{k=0}^{K_\delta} N_k > \mathcal{T}_\delta(K_\delta) + \frac{K_\delta + 1 + 4\log(1/\eta)}{2\log(1/\alpha)}\right) \le \eta. \tag{15}$$

Proof. We first strengthen Theorem 4.1 to an exponential moment bound. Fix outer iteration k and $0 < \theta < \log(1/\alpha)$. We define

$$l_k = \frac{(\log\beta_k)_+}{s\log(1/\alpha)}, \qquad A_\theta = \frac{\exp(s\theta) - 1}{1 - \alpha^s\exp(s\theta)}.$$

Since $\beta_k \le \exp(sl_k\log(1/\alpha))$ and N_k takes values in $\{0, s, 2s, \dots\}$, summing its conditional tails gives

$$\begin{aligned}\mathbf{E}_k[\exp(\theta N_k)] &= 1 + (\exp(s\theta) - 1)\sum_{j=0}^{\infty}\exp(s\theta j)\mathbf{P}_k(N_k > sj) \\ &\le \exp(s\theta l_k)\left(1 + A_\theta\alpha^{s\lceil l_k\rceil - l_k}\right).\end{aligned}$$

The function $d \mapsto \exp(s\theta d)(1 + A_\theta\alpha^{sd})$ is a sum of two exponentials and is convex on $[0,1]$. Its values at both endpoints equal $\frac{(1-\alpha^s)\exp(s\theta)}{1-\alpha^s\exp(s\theta)}$. Consequently,

$$\mathbf{E}_k[\exp(\theta(N_k - sl_k))] \le \frac{(1-\alpha^s)\exp(s\theta)}{1-\alpha^s\exp(s\theta)}.$$

Let us define

$$\Gamma(\theta) := \log\left(\frac{(1-\alpha^s)\exp(s\theta)}{1-\alpha^s\exp(s\theta)}\right) - \frac{s\theta}{1-\alpha^s}.$$

Then $\Gamma(0) = \Gamma'(0) = 0$, and

$$\Gamma''(\theta) = \frac{s^2\alpha^s\exp(s\theta)}{(1-\alpha^s\exp(s\theta))^2} = \frac{s^2}{4\sinh^2(s(\log(1/\alpha) - \theta)/2)} \le \frac{1}{(\log(1/\alpha) - \theta)^2}.$$

Integrating twice and expanding the logarithm yields

$$\begin{aligned}\log\mathbf{E}_k\left[\exp\left(\theta\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right)\right)\right] &\le -\log\left(1 - \frac{\theta}{\log(1/\alpha)}\right) - \frac{\theta}{\log(1/\alpha)} \\ &\le \frac{\theta^2}{2\log(1/\alpha)^2(1 - \frac{\theta}{\log(1/\alpha)})} =: \Psi(\theta).\end{aligned} \tag{16}$$

To account for termination, let $I_k = \mathbf{1}_{\{K_\delta \geq k\}}$ and $Y_k = \left(N_k - sl_k - \frac{s}{1-\alpha^s}\right)\mathbf{1}_{\{K_\delta \geq k\}}$, setting $N_k = sl_k = 0$ after termination. The indicator $\mathbf{1}_{\{K_\delta \geq k\}}$ is $\mathcal{F}_k$-measurable. By (16), the process

$$M_j = \exp\left(\sum_{k=0}^{j-1}\left(\theta\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right) - \Psi(\theta)\right)\mathbf{1}_{\{K_\delta \geq k\}}\right), \qquad 0 \leq j \leq K_\delta^+ + 1,$$

is a nonnegative supermartingale with $M_0 = 1$. Thus, $\mathbf{E}[M_j] \leq 1$ for all j. Note that $\sum_{k=0}^{K_\delta^+} \mathbf{1}_{\{K_\delta \geq k\}} = K_\delta + 1 \leq K_\delta^+ + 1$, so

$$\begin{aligned}\mathbf{E}\left[\exp\left(\theta\sum_{k=0}^{K_\delta}\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right)\right)\right] &= \mathbf{E}\left[\exp\left(\theta\sum_{k=0}^{K_\delta^+}\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right)\mathbf{1}_{\{K_\delta \geq k\}}\right)\right] \\ &\leq \exp((K_\delta^+ + 1)\Psi(\theta)).\end{aligned}$$

Markov's inequality, with

$$\theta = \frac{\log(1/\alpha)\sqrt{2\log(1/\eta)}}{\sqrt{K_\delta^+ + 1} + \sqrt{2\log(1/\eta)}},$$

gives

$$\mathbf{P}\left(\sum_{k=0}^{K_\delta}\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right) > \frac{\sqrt{2(K_\delta^+ + 1)\log(1/\eta) + \log(1/\eta)}}{\log(1/\alpha)}\right) \leq \exp(-\log(1/\eta)).$$

The pathwise telescoping bound (13) of Theorem 4.2 gives

$$\begin{aligned}\sum_{k=0}^{K_\delta} sl_k &\leq \frac{K_\delta \log D_\sigma}{\log(1/\alpha)} + \log\frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2\log(1/\alpha)} \\ \sum_{k=0}^{K_\delta} N_k &\leq \mathcal{T}_\delta(K_\delta) + \sum_{k=0}^{K_\delta}\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right).\end{aligned}$$

Combining the previous inequalities shows that $\sum_{k=0}^{K_\delta}\left(sl_k + \frac{s}{1-\alpha^s}\right) \leq \mathcal{T}_\delta(K_\delta)$, proving (14).

Alternatively, applying Markov's inequality directly to $M_{K_\delta^+ + 1}$ gives, for every fixed $0 < \theta < \log(1/\alpha)$,

$$\mathbf{P}\left(\sum_{k=0}^{K_\delta}\left(N_k - sl_k - \frac{s}{1-\alpha^s}\right) > \frac{(K_\delta + 1)\Psi(\theta) + \log(1/\eta)}{\theta}\right) \leq \exp(-\log(1/\eta)).$$

Choosing $\theta = \frac{1}{2}\log(1/\alpha)$, for which $\Psi(\theta) = \frac{1}{4}$, proves (15). □

Remark 4.2 (Concentration of the outer stopping time). Replacing $\mathcal{T}_\delta(K_\delta)$ by $\mathcal{T}_\delta(K_\delta^+)$ in (14) gives a deterministic upper bound with confidence $1 - \eta$. However, replacing K_δ by $\mathbf{E}[K_\delta]$ in the probability bounds (14) and (15) would require additional control of the outer stopping time. Since $\mathcal{T}_\delta$ is an increasing affine function, a bound on the upper tail of $K_\delta - \mathbf{E}[K_\delta]$ could be combined with

Theorem 4.3 by a union bound. The resulting work bound would involve $\mathcal{T}_\delta(\mathbf{E}[K_\delta])$ and an additional deviation term.

Certain parameterized families of problems and inner solvers may admit sharper results with the aforementioned upper tail decay. Establishing concentration about $\mathbf{E}[K_\delta]$ over popular classes of problems and applying it to practical algorithms is an interesting avenue for future work.

Under the assumptions of Theorem 3.2, $\mathbf{E}[K_\delta] = O(\log(1/\delta))$, so the amount of total inner work is $O(\log(1/\delta))$. This highlights the advantage of having a bound in terms of $\mathbf{E}[K_\delta]$, as we did not need to redo any of the outer convergence analysis. An interesting related question is to see if this can be used to formulate an average-case inner work complexity, rather than the worst-case complexity presented here. We leave this direction for future work.

Theorem 4.4 (A deterministic bound)**.** *Suppose we have a deterministic guarantee on the inner procedure of the form:*

$$\|\gamma z^l + y^l - w_k\|^2 \le C_0 \alpha^l \|\gamma z^0 + y^0 - w_k\|^2,$$

where the constants are the same as in Assumption 4.1. Let $D = \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2}$*, as before, and define*

$$\tilde{\beta}_k = \frac{(1+\sigma)^2 C_0 \|\gamma z^0 + y^0 - w_k\|^2}{\sigma^2 \max\{\|R(w_k)\|^2, \gamma^2\delta^2\}},$$

where (y^0, z^0) *is the initial pair of the* k*th inner solve. If* $\sigma \in (0,1)$ *and for* $k \ge 1$ *the inner solver is warm-started as in Lemma 4.1, then every inner solve terminates, with*

$$N_k \le s \left\lceil \frac{(\log \tilde{\beta}_k)_+}{s \log(1/\alpha)} \right\rceil \qquad (0 \le k \le K_\delta), \tag{17}$$

and the total inner work satisfies

$$\sum_{k=0}^{K_\delta} N_k \le s \left\lceil \frac{(\log \tilde{\beta}_0)_+}{s \log(1/\alpha)} \right\rceil + K_\delta \left(s + \frac{\log D_\sigma}{\log(1/\alpha)} \right) + \frac{1}{\log(1/\alpha)} \log \frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}. \tag{18}$$

Proof. Fix an outer iteration k and a graph pair (y, z) with residual $e = \gamma z + y - w_k$. The proof of Theorem 4.1 shows that if both tests of Algorithm 1 fail at (y, z), then $\|e\| > \frac{\sigma}{1+\sigma} \max\{\|R(w_k)\|, \gamma\delta\}$. That argument used only (4) and the triangle inequality, so no randomness is involved. Hence one of the tests passes as soon as $\|e\| \le \frac{\sigma}{1+\sigma} \max\{\|R(w_k)\|, \gamma\delta\}$.

At the check after sl inner iterations, the deterministic guarantee and an argument similar to Theorem 4.1 give

$$\begin{aligned} \|\gamma z^{sl} + y^{sl} - w_k\|^2 &\le C_0 \alpha^{sl} \|\gamma z^0 + y^0 - w_k\|^2 \\ &= \tilde{\beta}_k \alpha^{sl} \left(\frac{\sigma}{1+\sigma} \right)^2 \max\{\|R(w_k)\|, \gamma\delta\}^2. \end{aligned}$$

By the first paragraph, the inner loop exits by the first check l with $\tilde{\beta}_k \alpha^{sl} \le 1$, and $l \ge \lceil \frac{(\log \tilde{\beta}_k)_+}{s \log(1/\alpha)} \rceil$ satisfies this. This proves (17), and in particular $N_k \le s + \frac{(\log \tilde{\beta}_k)_+}{\log(1/\alpha)}$.

For $1 \le k \le K_\delta$, the inequalities of Lemma 4.1 and Corollary 4.1.1 also hold deterministically, with $\tilde{\beta}_k$ in place of β_k. The argument in the proof of Theorem 4.2 therefore gives the analog of (13)

with $K = K_\delta$:

$$\sum_{k=1}^{K_\delta} N_k \le K_\delta s + \frac{1}{\log(1/\alpha)} \sum_{k=1}^{K_\delta} (\log \tilde{\beta}_k)_+$$
$$\le K_\delta \left(s + \frac{\log D_\sigma}{\log(1/\alpha)} \right) + \frac{1}{\log(1/\alpha)} \log \frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}.$$

Adding the bound (17) for $k = 0$ gives the desired inequality in (18). □

Remark 4.3. Compared with Theorem 4.2, the per-iteration constant $\frac{s}{1-\alpha^s}$ improves to s, because there is no random tail to sum. Remark B.1 in Appendix A shows that the per-step bound (17) is attained, that the expectation bound of Theorem 4.1 can be approached arbitrarily closely, and that, for $\sigma \le \sigma_1$, the constant D_σ in the total bounds is sharp up to the factor $(1+\sigma)^2$.

One might wonder if Assumption 4.1 is strong enough to ensure that every inner loop terminates when the terminal test of Algorithm 1 is removed, i.e., when only the relative test is used. We show with a simple example that this is not the case. This phenomenon has already been noted in Eckstein and Yao [2018].

Example 4.1 (An Infinite Inner Loop). Suppose $\mathcal{H} = \mathcal{D} = \mathbb{R}$, $C = 1$, $\gamma = 1$, and $A = I$. This is equivalent to minimizing $f(x) = \frac{1}{2}x^2$. A valid graph pair has $z = u = y$ and error $e = z + y - w = 2u - w$. At the exact outer fixed point $w = 0$, take $u^l = z^l = \alpha^{l/2}$ for $0 < \alpha < 1$, so $u^0 = 1$ (superscripts denote inner iterations, as throughout). Then $|e^l|^2 = 4\alpha^l = \alpha^l |e^0|^2$, but

$$|e^l| = 2|u^l| > \sigma |u^l - w| \qquad (0 < \sigma < 1)$$

at every finite index. Consequently a purely relative inner loop never terminates, even under deterministic geometric contraction. The terminal condition $|z^l| = \alpha^{l/2} \le \delta$ does terminate for every $\delta > 0$. This shows why merely declaring the outer point to be a fixed point, or invoking convergence of e^l to zero, does not justify finite inner termination.

4.2 Selection of Parameters

We now use Theorem 4.2 to choose the stride s, given the costs of an inner iteration and of a check. The choice of σ depends on the inner solver and is examined for specific methods in Propositions 5.1 and 5.3.

Lemma 4.2. *Suppose the cost of testing is c_{test}, the cost of each inner iteration is c_{in}, the cost of each outer iteration is c_{out}, and the cost of initialization is c_{init}. Given an upper bound $K_{\text{up}} \ge \mathbf{E}[K_\delta]$, under the assumptions of Theorem 4.2, we can bound the total cost M as*

$$\mathbf{E}[M] \le c_{\text{init}} + c_{\text{out}}(K_{\text{up}} + 1) + c_{\text{in}} N_{\text{tot}} + c_{\text{test}}(K_{\text{up}} + 1 + s^{-1} N_{\text{tot}}),$$

where

$$N_{\text{tot}} := (K_{\text{up}} + 1)\left(s + \frac{1}{\log(1/\alpha)} \right) + N',$$
$$N' := \frac{1}{\log(1/\alpha)} \left(K_{\text{up}} \log D_\sigma + (\log \beta_0)_+ + \log \frac{\max\left\{\|R(w_0)\|^2, \gamma^2\delta^2\right\}}{\gamma^2\delta^2} \right),$$

and D, β_0 are as in Theorem 4.2. The choice of stride $s \in \mathbb{N}$ that minimizes this upper bound is one of $\lfloor s^\star \rfloor$ or $\lceil s^\star \rceil$, where

$$s^\star := \sqrt{\frac{c_{\text{test}}\left(N' + \frac{K_{\text{up}}+1}{\log(1/\alpha)}\right)}{c_{\text{in}}(K_{\text{up}}+1)}}.$$

Proof. Since $\frac{s}{1-\alpha^s} \le s + \frac{1}{\log(1/\alpha)}$ and $\mathbf{E}[K_\delta] \le K_{\text{up}}$, Theorem 4.2 gives $\mathbf{E}[\sum_{k=0}^{K_\delta} N_k] \le N_{\text{tot}}$. Outer iteration k performs one test before any inner iteration and one after every s inner iterations, i.e., $1 + N_k/s$ tests. There are $K_\delta + 1$ outer iterations, so

$$\begin{aligned}
\mathbf{E}[M] &= \mathbf{E}\left[c_{\text{init}} + \sum_{k=0}^{K_\delta}\left[c_{\text{out}} + c_{\text{in}}N_k + c_{\text{test}}(1 + s^{-1}N_k)\right]\right] \\
&\le c_{\text{init}} + c_{\text{out}}(K_{\text{up}}+1) + c_{\text{in}}N_{\text{tot}} + c_{\text{test}}(K_{\text{up}} + 1 + s^{-1}N_{\text{tot}}) \qquad (19) \\
&= M' + c_{\text{in}}(K_{\text{up}}+1)\,s + c_{\text{test}}\left(N' + \frac{K_{\text{up}}+1}{\log(1/\alpha)}\right)s^{-1},
\end{aligned}$$

where M' suppresses the terms not involving the stride s. Writing $M_1 := c_{\text{in}}(K_{\text{up}}+1)$ and $M_2 := c_{\text{test}}\left(N' + \frac{K_{\text{up}}+1}{\log(1/\alpha)}\right)$, the right-hand side is $M' + M_1 s + M_2 s^{-1}$ with $M', M_1, M_2 \ge 0$. As a function of s, it is convex on $(0,\infty)$ and minimized at $s^\star = \sqrt{M_2/M_1}$. Since it is decreasing on $(0, s^\star]$ and increasing on $[s^\star, \infty)$, its minimizer over the positive integers is one of $\lfloor s^\star \rfloor$ or $\lceil s^\star \rceil$. □

Remark 4.4. The only relaxation in this bound beyond Theorem 4.2 is the use of the inequality $\frac{s}{1-\alpha^s} \le s + \frac{1}{\log(1/\alpha)}$, which costs at most $\frac{s}{2}$ inner iterations per outer iteration. It is a heuristic that one could reasonably estimate before launching their algorithm of choice. For an exact answer, one would replace $s + \frac{1}{\log(1/\alpha)}$ in N_{tot} in (19) with $\frac{s}{1-\alpha^s}$ and attempt to numerically find the minimum.

5 Application to First-Order Methods

In general, the relationship between parameters is complex and depends on a variety of factors. Moreover, each choice of algorithm requires a separate analysis of how to optimally choose the parameters σ, s, and γ. In this section, we consider a few common methods and propose heuristics for selecting parameters. Note that these heuristics are oriented toward reduction of the fixed point residual, which may not be the metric of interest in all cases. Nevertheless, we hope this provides a more principled heuristic for implementations of these algorithms.

5.1 Proximal Point Method

As a prescriptive example, let us consider the case of inexact proximal point on a smooth convex function. The optimal choice of fixed stepsize is $\gamma_{\text{GD}} = \frac{2}{(L+\frac{1}{\gamma})+\frac{1}{\gamma}} = \frac{2\gamma}{\gamma L+2}$, where γ_{GD} is the inner stepsize of the gradient method applied to the proximal subproblem $\min_x \left\{f(x) + \frac{1}{2\gamma}\|x-w\|^2\right\}$, which is $(L+\frac{1}{\gamma})$-smooth and $(\frac{1}{\gamma})$-strongly convex.

Proposition 5.1. *Consider the case where $C = I$ and $A = \nabla f$ where f is convex and L-smooth. At outer iteration k, suppose the inner procedure is gradient descent with stepsize $\gamma_{\text{GD}} = \frac{2\gamma}{\gamma L+2}$ on the proximal subproblem $\operatorname{argmin}_x \varphi_k(x)$, where $\varphi_k(x) := f(x) + \frac{1}{2\gamma}\|x - w_k\|^2$, warm-started at the previously accepted u, and uses the graph pair $(u^l, z^l) = (x^l, \nabla f(x^l))$. Then, the following hold:*

1. *The deterministic guarantee of Theorem 4.4, and hence Assumption 4.1, holds with*

$$\alpha = \left(\frac{\gamma L}{\gamma L + 2}\right)^2, \qquad C_0 = (1+\gamma L)^2.$$

2. *Let* $\chi := \frac{s\log(1/\alpha)}{1-\alpha^s}$ *for the bound of Theorem 4.2, or* $\chi := s\log(1/\alpha)$ *for the bound of Theorem 4.4. Substitute* $\mathbf{E}[K_\delta] \le K_\delta^+ \le \frac{1+\sigma}{1-\sigma}\cdot\frac{\mathbf{dist}(w_0,\mathbf{zer}\,B)^2}{\gamma^2\delta^2}$ *into either bound. The coefficient of* δ^{-2} *is then* $\frac{\mathbf{dist}(w_0,\mathbf{zer}\,B)^2}{\gamma^2\log(1/\alpha)}g(\sigma)$*, where*

$$g(\sigma) := \frac{1+\sigma}{1-\sigma}\left(\chi + \log D_\sigma(\sigma)\right), \qquad D(\sigma) = \frac{(1+\sigma)^2 C_0 G_\sigma^2}{\sigma^2}.$$

Over $\sigma \in (0,1)$*,* g *has a unique minimizer* $\sigma^\star$*, which lies in* $(0,\sigma_1)$ *and is the unique solution in* $(0,\sigma_1)$ *of*

$$\frac{1}{\sigma} = 1 + \psi(\sigma) + \chi + \log\frac{(1+\sigma)^2(1+2\sigma)^2 C_0}{\sigma^2}, \qquad \psi(\sigma) := \frac{2(1-\sigma^2)}{1+2\sigma}. \tag{20}$$

Moreover,

$$\sigma^\star < \frac{1}{3/2 + s\log(1/\alpha) + 2\log(1+\gamma L)}.$$

The other σ*-dependent term in the bounds,* $\frac{(\log\beta_0)_+}{\log(1/\alpha)}$*, does not grow as* $\delta \to 0$*. Consequently,* $\sigma^\star$ *minimizes the upper bound to leading order in* δ^{-2}*.*

Proof. Here $B = \nabla f$, $y = u$, and $J = (I + \gamma\nabla f)^{-1}$ maps w_k to the minimizer $x_k^\star$ of φ_k. The pair $(x^l, \nabla f(x^l))$ lies in $\mathbf{gra}\,B$, and its residual is $e^l = \gamma\nabla f(x^l) + x^l - w_k = \gamma\nabla\varphi_k(x^l)$. The function φ_k is μ-strongly convex and L_φ-smooth with $\mu = \frac{1}{\gamma}$ and $L_\varphi = L + \frac{1}{\gamma}$, so its condition number is $\frac{L_\varphi}{\mu} = 1 + \gamma L$. With stepsize $\frac{2}{L_\varphi+\mu} = \gamma_{\mathrm{GD}}$, gradient descent satisfies $\|x^l - x_k^\star\| \le \alpha^{l/2}\|x^0 - x_k^\star\|$ with $\alpha = \left(\frac{L_\varphi-\mu}{L_\varphi+\mu}\right)^2 = \left(\frac{\gamma L}{\gamma L+2}\right)^2$ [Nesterov, 2013, Theorem 2.1.15]. Since $\mu\|x - x_k^\star\| \le \|\nabla\varphi_k(x)\| \le L_\varphi\|x - x_k^\star\|$,

$$\begin{aligned}
\|e^l\|^2 &= \gamma^2\|\nabla\varphi_k(x^l)\|^2 \\
&\le \gamma^2 L_\varphi^2\alpha^l\|x^0 - x_k^\star\|^2 \\
&\le \left(\frac{L_\varphi}{\mu}\right)^2\alpha^l\gamma^2\|\nabla\varphi_k(x^0)\|^2 \\
&= (1+\gamma L)^2\alpha^l\|e^0\|^2.
\end{aligned}$$

This is a deterministic bound that holds for every inner iteration l, independently of outer iteration k, so we are in the setting of Theorem 4.4.

Define $S = \frac{1+\sigma}{1-\sigma}$ and $\Delta = \frac{\mathbf{dist}(w_0,\mathbf{zer}\,B)^2}{\gamma^2\delta^2}$. The bound of Theorem 4.2 is at most

$$\frac{S\Delta}{\log(1/\alpha)}(\chi + \log D_\sigma(\sigma)) + \frac{s}{1-\alpha^s} + \frac{1}{\log(1/\alpha)}\left((\log\beta_0)_+ + \log\frac{\max\{\|R(w_0)\|^2, \gamma^2\delta^2\}}{\gamma^2\delta^2}\right).$$

The bound of Theorem 4.4 has the same δ^{-2} term with $\chi = s\log(1/\alpha)$, and its remaining terms also do not grow with δ^{-1} faster than $\log(1/\delta)$. This identifies the δ^{-2} coefficient as claimed. Here α, C_0, and hence χ do not depend on σ.

Let $\tilde{D}(\sigma) := \frac{(1+\sigma)^2(1+2\sigma)^2 C_0}{\sigma^2}$ and $\tilde{g}(\sigma) := S(\sigma)\left(\chi + \log \tilde{D}(\sigma)\right)$. By Lemma 4.1, $G_\sigma \geq 1 + 2\sigma$ with equality for $\sigma \leq \sigma_1$, so $g \geq \tilde{g}$ on $(0,1)$, with equality on $(0, \sigma_1]$. We first minimize $\tilde{g}$. Since $\log \tilde{D} = \log C_0 + 2\log(1+\sigma) + 2\log(1+2\sigma) - 2\log\sigma$,

$$\frac{1-\sigma^2}{2}\frac{d}{d\sigma}\log \tilde{D} = 1 + \psi(\sigma) - \frac{1}{\sigma},$$

with $\psi(\sigma) = \frac{2(1-\sigma^2)}{1+2\sigma}$ as in the optimality condition (20). Using $\frac{d}{d\sigma}\log S = \frac{2}{1-\sigma^2}$, we get

$$\begin{aligned}
\tilde{g}'(\sigma) &= \frac{2S}{1-\sigma^2}\left(\chi + \log \tilde{D}(\sigma) + 1 + \psi(\sigma) - \frac{1}{\sigma}\right) \\
&= -\frac{2S}{1-\sigma^2}\, h(\sigma), \\
h(\sigma) &:= \frac{1}{\sigma} - 1 - \psi(\sigma) - \chi - \log \tilde{D}(\sigma).
\end{aligned}$$

Now $h(\sigma) \to +\infty$ as $\sigma \downarrow 0$, since $1/\sigma$ dominates $\log \tilde{D}(\sigma) = 2\log(1/\sigma) + O(1)$, and $h(1) = -\chi - \log(36 C_0) < 0$. Moreover, h is strictly decreasing: since $\psi'(\sigma) = -\frac{4(\sigma^2+\sigma+1)}{(1+2\sigma)^2}$,

$$h'(\sigma) = -\frac{1}{\sigma^2} + \frac{2}{\sigma} - \frac{2}{1+\sigma} - \frac{4}{1+2\sigma} + \frac{4(\sigma^2+\sigma+1)}{(1+2\sigma)^2} = -\frac{(1-\sigma)^2}{\sigma^2} - \frac{1-\sigma}{1+\sigma} - \frac{4\sigma(1-\sigma)}{(1+2\sigma)^2} < 0.$$

Hence h has a unique root $\sigma^\star \in (0,1)$, which solves (20), and $\tilde{g}$ is strictly decreasing on $(0, \sigma^\star)$ and strictly increasing on $(\sigma^\star, 1)$. Furthermore $\sigma^\star < \sigma_1$, because $h(\sigma_1) < 0$: indeed $\psi(\sigma_1) = \frac{1}{2}$ and $\frac{1}{\sigma_1} - 1 - \psi(\sigma_1) = \frac{1+\sqrt{13}}{3} - \frac{3}{2} < 0.04$, while $\log \tilde{D}(\sigma_1) = \log C_0 + 2\log\frac{17+5\sqrt{13}}{6} > \log C_0 + 2\log 5 > 3$. It follows that $\sigma^\star$ is also the unique minimizer of g: for $\sigma \leq \sigma_1$, $g(\sigma) = \tilde{g}(\sigma) \geq \tilde{g}(\sigma^\star) = g(\sigma^\star)$ with equality only at $\sigma = \sigma^\star$, and for $\sigma > \sigma_1$, $g(\sigma) \geq \tilde{g}(\sigma) > \tilde{g}(\sigma_1) \geq \tilde{g}(\sigma^\star) = g(\sigma^\star)$. Finally, $\log \tilde{D}(\sigma) > \log C_0$ on $(0,1)$, and ψ is decreasing, so $\psi(\sigma^\star) \geq \psi(\sigma_1) = \frac{1}{2}$. Substituting into (20) gives $\frac{1}{\sigma^\star} > \frac{3}{2} + \chi + \log C_0$. □

Proposition 5.2. *Consider the setting of Proposition 5.1, with $L > 0$. Define $\kappa := 1 + \gamma L$, $L_\varphi := L + \frac{1}{\gamma}$, and $\rho := \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$. At each outer iteration, apply Nesterov's accelerated gradient method to φ_k, with x^0 warm-started at the previously accepted u and $x^{-1} = x^0$:*

$$\begin{aligned}
v^l &= x^l + \rho(x^l - x^{l-1}), \\
x^{l+1} &= v^l - \frac{1}{L_\varphi}\nabla\varphi_k(v^l), \qquad l = 0, 1, 2, \ldots.
\end{aligned}$$

Note that momentum is reset at the beginning of each inner solve. Use the graph pair $(u^l, z^l) = (x^l, \nabla f(x^l))$ at each stopping test. Then the following hold:

1. *The deterministic guarantee of Theorem 4.4, and hence Assumption 4.1, holds with*

$$\alpha = 1 - \frac{1}{\sqrt{\kappa}}, \qquad C_0 = 2\kappa. \tag{21}$$

2. *Let $\chi := \frac{s\log(1/\alpha)}{1-\alpha^s}$ for the expectation bound, or $\chi := s\log(1/\alpha)$ for the deterministic bound. After substituting the outer iteration bound as in Proposition 5.1, the coefficient of δ^{-2} is*

$$\frac{\mathbf{dist}(w_0, \mathbf{zer}\, B)^2}{\gamma^2 \log(1/\alpha)} g(\sigma), \qquad g(\sigma) := \frac{1+\sigma}{1-\sigma}\left(\chi + \log\frac{2\kappa(1+\sigma)^2 G_\sigma^2}{\sigma^2}\right).$$

The function g has a unique minimizer $\sigma^\star \in (0, \sigma_1)$, where $\sigma_1 = \frac{\sqrt{13}-1}{4}$. It is the unique solution of (20) in this interval with $C_0 = 2\kappa$ and the above choice of χ. Moreover,

$$\sigma^\star < \frac{1}{3/2 + \chi + \log(2\kappa)} \leq \frac{1}{3/2 + s\log(1/\alpha) + \log(2\kappa)}.$$

For fixed initialization and algorithm parameters, the remaining terms grow at most as $\log(1/\delta)$ as $\delta \downarrow 0$. Consequently, $\sigma^\star$ minimizes the upper bound to leading order in δ^{-2}.

The proof is given in Appendix C, as it is extremely similar to the proof of Proposition 5.1.

Note that the inner iterations of Proposition 5.1 are of the form $x_k^{l+1} = x_k^l - \gamma_{\mathrm{GD}}(\nabla f(x_k^l) + \frac{1}{\gamma}(x_k^l - w_k))$. That is, all future iterates are in the span of past iterates and gradients. Any such method needs $\sum_{k=0}^{K_\delta} N_k \gtrsim \delta^{-1/2}$. This follows[2] from analyzing "the worst function in the world" [Nesterov, 2013, Section 2.1.2]. One can see a similar property for Proposition 5.2. We leave a more detailed analysis of lower bounds future work.

5.2 Douglas–Rachford

Alves et al. [2026] show that the inexact Douglas–Rachford splitting of Eckstein and Yao [2018] fits into their framework. We reproduce the formulation here. Suppose we wish to solve

$$\text{find } x \text{ such that } 0 \in A_1 x + A_2 x, \tag{22}$$

where A_1 and A_2 are maximal monotone operators. We introduce the operators

$$A = \begin{bmatrix} \tau A_1 & I & -I \\ -I & \tau A_2 & I \\ I & -I & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} I \\ -I \\ I \end{bmatrix}, \qquad M = CC^*,$$

and set $\gamma = 1$. One can check that $(CC^* + \gamma A)^{-1} CC^*$ is defined everywhere. Note that if $0 \in A(x_1, x_2, v)$, then $x_1 = x_2 =: x$ and $0 \in A_1 x + A_2 x$. Substituting this into Algorithm 1 yields Algorithm 2. We also remark that this is not the same formulation as that of Eckstein and Bertsekas [1992], who show that Douglas–Rachford splitting is an instance of PPM for a certain splitting operator. However, the splitting operator is difficult to define and understand. This formulation shows that Douglas–Rachford splitting can be viewed as the projection of PPM done in a product space.

In the notation of Algorithm 1 (writing its block variable as (x_1, x_2, v) to avoid a clash with $u_k \in A_1 x_k$), the graph pair produced in lines 3–4 has $z_k = x_k - y_k$ and $C^*(x_1, x_2, v) = y_k + \tau u_k$, so that $\gamma z_k + C^*(x_1, x_2, v) - w_k = x_k + \tau u_k - w_k$. Lines 5, 7, and 8 are therefore the terminal test, relative test, and update of Algorithm 1. At termination, the returned quadruple satisfies $u_k \in A_1 x_k$, $v_k \in A_2 y_k$, $\|x_k - y_k\| \leq \delta$, and, by line 4, $\|u_k + v_k\| = \frac{1}{\tau}\|x_k - y_k\| \leq \frac{\delta}{\tau}$. That is, (x_k, y_k) solves (22) up to δ in the primal variables and $\frac{\delta}{\tau}$ in the operator values.

5.2.1 ADMM

Eckstein and Yao [2018] also show how their formulation yields a partially inexact version of ADMM. Suppose we want to solve

$$\begin{array}{ll} \text{minimize} & f(x) + g(z) \\ \text{subject to} & x = z, \end{array} \tag{23}$$

[2]Namely, using Theorem 3.1 to conclude that gradients and iterates are bounded by initialization constants.

Algorithm 2 The method introduced in Eckstein and Yao [2018] (adapted from [Alves et al., 2026, Algorithm 2]) as an instance of Algorithm 1.

1: **Initialize:** $w_0 \in \mathcal{H}$, stepsize $\tau > 0$, inexactness $\sigma \in [0, 1)$, stride $s \in \mathbb{N}$, and stopping tolerance δ
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Produce a pair $(x_k, u_k) \in \mathcal{H} \times \mathcal{H}$ with

$$u_k \in A_1 x_k, \qquad \tau u_k + x_k \approx w_k.$$

4: Compute $y_k = J_{\tau A_2}(x_k - \tau u_k)$ and $v_k = \frac{1}{\tau}(x_k - \tau u_k - y_k)$, so that

$$v_k \in A_2 y_k, \qquad \tau v_k + y_k = x_k - \tau u_k.$$

5: **if** $\|x_k - y_k\| \le \delta$ **then**
6: Stop and return (x_k, y_k, u_k, v_k).
7: **else if** $\|x_k + \tau u_k - w_k\| \le \sigma \|y_k + \tau u_k - w_k\|$ **then**
8: Set $w_{k+1} = w_k - \tau(u_k + v_k)$ (equivalently, $w_{k+1} = w_k + y_k - x_k$).
9: **else**
10: Improve the pair (x_k, u_k) with $u_k \in A_1 x_k$ to reduce $\|x_k + \tau u_k - w_k\|$ via the iterative procedure s times and go to line 4.
11: **end if**
12: **end for**

where x and z are the optimization variables, and f and g are closed, proper, and convex functions, but not necessarily smooth. Its optimality condition is $0 \in \partial f(x) + \partial g(x)$, i.e., the inclusion (22) with $A_1 = \partial f$ and $A_2 = \partial g$, whenever $\partial(f + g) = \partial f + \partial g$.[3] Eckstein and Yao [2018] describe a variant of ADMM which produces iterates as

$$x_k \approx \operatorname*{argmin}_x f(x) + \langle p_k, x \rangle + \frac{c}{2}\|x - z_k\|^2, \tag{24a}$$

$$p_{k+1} = p_k + c(x_k - z_k), \tag{24b}$$

$$z_{k+1} = \operatorname*{argmin}_z g(z) - \langle p_{k+1}, z \rangle + \frac{c}{2}\|z - x_k\|^2. \tag{24c}$$

Here, as in Algorithms 1 and 2, the x-iterate carries the index of the outer iteration that produces it, so x_k is what is usually written x_{k+1} in ADMM. The state carried between outer iterations is (z_k, p_k). We reproduce their algorithm within the framework established by Alves et al. [2026]. Take $A_1 = \partial f$, $A_2 = \partial g$, and $\tau = \frac{1}{c}$ in the Douglas–Rachford instance above, that is,

$$A = \begin{bmatrix} \frac{1}{c}\partial f & I & -I \\ -I & \frac{1}{c}\partial g & I \\ I & -I & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} I \\ -I \\ I \end{bmatrix}, \qquad \gamma = 1, \tag{25}$$

and change variables by $w_k = z_k - \frac{p_k}{c}$. The adjoint operator C^* is clearly surjective. Also, $0 \in A(x_1, x_2, v)$ if and only if $x_1 = x_2 =: x$ and $v - x \in \frac{1}{c}\partial f(x)$, $x - v \in \frac{1}{c}\partial g(x)$. Hence $\mathbf{zer}\, A \neq \varnothing$ if and only if $0 \in \partial f(x) + \partial g(x)$ has a solution.

[3] Note that we always have $\partial f(x) + \partial g(x) \subseteq \partial(f + g)(x)$ [Bauschke and Combettes, 2017, Proposition 16.6].

Only the f-subproblem is solved inexactly. The inner solver returns a pair (x_k, a_k) with $a_k \in \partial f(x_k)$. The g-subproblem is solved exactly. Writing $\varphi_k(x) := f(x) + \langle p_k, x\rangle + \frac{c}{2}\|x - z_k\|^2$ for the x-subproblem, the graph residual of Algorithm 1 becomes

$$x_k + \frac{a_k}{c} - w_k = \frac{1}{c}(a_k + p_k + c(x_k - z_k)) \in \frac{1}{c}\partial\varphi_k(x_k). \tag{26}$$

That is, it is a scaled subgradient of the x-subproblem objective, which is zero exactly when x_k solves the subproblem with $a_k = -p_{k+1}$. Substituting this formulation into Algorithm 1 yields Algorithm 3. We formally prove this in Proposition 5.3.

Algorithm 3 Inexact ADMM of Eckstein and Yao [2018] as an instance of Algorithm 1.

1: **Initialize:** z_0, p_0 (equivalently $w_0 = z_0 - \frac{p_0}{c}$), penalty $c > 0$, inexactness $\sigma \in [0, 1)$, stride $s \in \mathbb{N}$, and stopping tolerance δ
2: **for** $k = 0, 1, 2, \ldots$ **do**
3: Produce (x_k, a_k) with $a_k \in \partial f(x_k)$ and $x_k \approx \operatorname{argmin}_x \varphi_k(x)$, warm-started at (x_{k-1}, a_{k-1}) for $k \geq 1$.
4: Compute $z_{k+1} = \operatorname{argmin}_z \left\{g(z) + \langle a_k, z\rangle + \frac{c}{2}\|z - x_k\|^2\right\} = \operatorname{prox}_{g/c}\left(x_k - \frac{a_k}{c}\right)$.
5: **if** $\|x_k - z_{k+1}\| \leq \delta$ **then**
6: Stop and return (x_k, z_{k+1}, a_k).
7: **else if** $\|a_k + p_k + c(x_k - z_k)\| \leq \sigma\|a_k + p_k + c(z_{k+1} - z_k)\|$ **then**
8: Set $p_{k+1} = p_k + c(x_k - z_k)$ and keep z_{k+1}.
9: **else**
10: Improve (x_k, a_k) with $a_k \in \partial f(x_k)$ via the iterative procedure s times and go to line 4.
11: **end if**
12: **end for**

When the x-subproblem is solved exactly, $a_k = -p_{k+1}$, and lines 4 and 8 reduce to the ADMM iterates (24). Each check (lines 5 and 7 of Algorithm 3) costs one evaluation of $\operatorname{prox}_{g/c}$. The check uses the same output for both the stopping test and the relative error test. This is the test cost c_{test} in the optimal stride selection analysis of Lemma 4.2. The cost of c_{test} relative to the cost of one inner iteration c_{in} depends on the hardware and implementation details, but typically $c_{\text{test}} \gg c_{\text{in}}$ in practice.

Proposition 5.3. *Let A and C be the associated inexact PPM operators for ADMM (25), and suppose $w_0 = z_0 - \frac{p_0}{c}$. Then, the following hold:*

1. *Algorithm 3 is Algorithm 1 applied to A and C with $\gamma = 1$, with $w_k = z_k - \frac{p_k}{c}$ for all k. Its terminal and relative tests are those of Algorithm 1. Hence Theorems 3.1, 4.1, 4.2 and 4.4 apply, with the warm start of line 3 playing the role of the warm start in Lemma 4.1.*

2. *At termination, $a_k \in \partial f(x_k)$ and $\tilde{a} := c(x_k - z_{k+1}) - a_k \in \partial g(z_{k+1})$, with*

$$\|x_k - z_{k+1}\| \leq \delta, \qquad \|a_k + \tilde{a}\| = c\|x_k - z_{k+1}\| \leq c\delta.$$

   That is, (x_k, z_{k+1}) satisfies the optimality conditions of (23) up to δ in primal feasibility and $c\delta$ in dual feasibility.

3. *If f is L-smooth and the inner procedure is gradient descent on φ_k with stepsize $\frac{2}{L+2c}$ and $a^l = \nabla f(x^l)$, then the deterministic guarantee of Theorem 4.4 holds with*

$$\alpha = \left(\frac{L}{L + 2c}\right)^2, \qquad C_0 = \left(1 + \frac{L}{c}\right)^2.$$

Proof. We prove each item separately.

1. First, write the variables of Algorithm 1 as (x_1, x_2, v), with ζ in place of its z, to avoid a clash with the ADMM variables z_k and a_k. Expanding $C\zeta \in A(x_1, x_2, v)$ row by row, the condition is equivalent to

$$\begin{aligned} c(\zeta - x_2 + v) &\in \partial f(x_1), \\ c(x_2 - v) &\in \partial g(x_2), \\ \zeta &= x_1 - x_2. \end{aligned}$$

   Given (x_k, a_k) with $a_k \in \partial f(x_k)$ and z_{k+1} from line 4, set

$$\begin{aligned} x_1 &= x_k, \\ x_2 &= z_{k+1}, \\ \zeta &= x_k - z_{k+1}, \\ v &= \frac{a_k}{c} - \zeta + z_{k+1}. \end{aligned}$$

   The first and third conditions hold by construction. The second reads $c(x_k - z_{k+1}) - a_k \in \partial g(z_{k+1})$, which is the optimality condition of the z-subproblem in line 4. So line 4 produces a graph pair of B, and conversely every graph pair arises this way from its first block. With $y = C^*(x_1, x_2, v) = x_1 - x_2 + v$ and $\gamma = 1$,

$$\begin{aligned} y - w_k &= x_k + \frac{a_k}{c} - w_k - \gamma\zeta \\ &= z_{k+1} + \frac{a_k}{c} - w_k. \end{aligned}$$

   Substituting $w_k = z_k - \frac{p_k}{c}$ and multiplying by c, the relative test of Algorithm 1 becomes line 7. Its terminal test $\|\zeta\| \le \delta$ becomes line 5. For the update, $w_{k+1} = w_k - \zeta = w_k - x_k + z_{k+1}$. Defining $p_{k+1} := c(z_{k+1} - w_{k+1})$ preserves $w_{k+1} = z_{k+1} - \frac{p_{k+1}}{c}$ and gives

$$p_{k+1} = c(x_k - w_k) = p_k + c(x_k - z_k),$$

   which is line 8. Finally, (x_1, x_2, v, ζ) is a function of (x_k, a_k) alone, so warm-starting the f-solver at (x_{k-1}, a_{k-1}) reproduces the previously accepted graph pair, as required in Lemma 4.1. Note also that the residual (26) depends only on the f-solver's output, so the contraction assumptions concern the f-subproblem alone.

2. Now, for the second item, $\tilde{a} \in \partial g(z_{k+1})$ is the second graph condition above, and $a_k + \tilde{a} = c(x_k - z_{k+1}) = c\zeta$. The bounds follow from the terminal test.

3. Finally, up to an additive constant, $\varphi_k(x) = f(x) + \frac{c}{2}\|x - w_k\|^2$, so φ_k is the proximal subproblem of Proposition 5.1 with γ replaced by $\frac{1}{c}$. By (26), the residual of the gradient descent iterate x^l is $\frac{1}{c}\nabla\varphi_k(x^l)$, which matches the residual $\gamma\nabla\varphi_k(x^l)$ in the proof of Proposition 5.1. The stepsize $\frac{2}{L+2c} = \frac{2\cdot\frac{1}{c}}{L\cdot\frac{1}{c}+2}$ also matches. That proof therefore gives the third item with γL replaced by $\frac{L}{c}$.

□

5.3 Inexact Davis–Yin

We can extend the Douglas–Rachford result to the more general Davis–Yin case [Davis and Yin, 2017], where we wish to solve

$$\text{find } x \text{ such that } 0 \in A_1 x + A_2 x + A_3 x, \tag{27}$$

where A_1 and A_2 are maximal monotone operators and A_3 is L_3^{-1}-cocoercive.[4] Let $\tau \in (0, 2/L_3)$ be the Davis–Yin stepsize and let

$$0 < \theta \leq 1 - \frac{\tau L_3}{4}$$

be the relaxation parameter. As in the Douglas–Rachford case, the stepsize of Algorithm 1 is $\gamma = 1$, and τ plays the role of the stepsize inside A. We define

$$A = \begin{bmatrix} \tau A_1 & I & -I \\ \tau A_3 - I & \tau A_2 & I \\ \theta I & -\theta I & 0 \end{bmatrix}, \qquad C = \begin{bmatrix} I \\ -I \\ I \end{bmatrix}, \qquad \gamma = 1. \tag{28}$$

If $A_3 = 0$, one may take $\theta = 1$ (formally $L_3 = 0$), and the operators (28) reduce to the Douglas–Rachford operators above. Unlike in the Douglas–Rachford case, A is *not* monotone in general: its linear part is not skew-adjoint when $\theta \neq 1$ or $A_3 \neq 0$ (for instance, with $A_1 = A_2 = 0$ and $A_3 = L_3 I$ on $\mathbb{R}$, the symmetric part of A has zero diagonal and nonzero off-diagonal entries, so it is indefinite). Hence Assumption 2.1 does not hold. However, the operator B of Definition 2.1 is still maximal monotone (Proposition 5.4), so we may use the weaker Assumption 2.2, and the results in Sections 3 and 4 still hold. Substituting this into Algorithm 1 yields Algorithm 4. We formalize this in Proposition 5.4.

In the notation of Algorithm 1, the quantities tested in lines 5 and 7 are $\|z_k\|$, $\|\gamma z_k + C^* u_k - w_k\|$, and $\|C^* u_k - w_k\|$, with $z_k = \theta(x_k - y_k)$ and $C^* u_k = (1-\theta)x_k + \theta y_k + \tau u_k$ (Proposition 5.4). Each check costs one evaluation of A_3 and one resolvent of A_2. When the A_1-step is exact, $\tau u_k = w_k - x_k$, and lines 4 and 8 become $y_k = J_{\tau A_2}(2x_k - w_k - \tau A_3 x_k)$ and $w_{k+1} = w_k + \theta(y_k - x_k)$, i.e., Davis–Yin splitting with relaxation parameter θ.

Proposition 5.4. *Let A_1 and A_2 be maximal monotone, let A_3 be L_3^{-1}-cocoercive, let $\tau \in (0, 2/L_3)$ and $\theta \in (0, 1 - \frac{\tau L_3}{4}]$, and let A and C be as in (28). The following hold:*

1. *C^* is surjective, and $0 \in A(x_1, x_2, v)$ if and only if $x_1 = x_2 =: x$, $v - x \in \tau A_1 x$, and $x - v - \tau A_3 x \in \tau A_2 x$. In particular, $\mathbf{zer}\, A \neq \varnothing$ if and only if (27) has a solution.*
2. *$(C^*(x_1, x_2, v), \zeta) \in \mathbf{gra}\, B$ if and only if there is $u \in A_1 x_1$ with*

$$x_2 = J_{\tau A_2}(x_1 - \tau u - \tau A_3 x_1), \qquad v = \tau u - \theta x_1 + (1+\theta) x_2, \qquad \zeta = \theta(x_1 - x_2). \tag{29}$$

In that case, $C^*(x_1, x_2, v) = (1-\theta)x_1 + \theta x_2 + \tau u$, and $\zeta + C^*(x_1, x_2, v) - w = x_1 + \tau u - w$ for every w. Consequently, Algorithm 4 is Algorithm 1 applied to the operators (28) with (x_k, y_k, u_k) in the roles of (x_1, x_2, u): lines 5, 7, and 8 are its terminal test, relative test, and update, and the warm start of Lemma 4.1 is the warm start of the A_1-solver at (x_{k-1}, u_{k-1}).

[4] That is, $\langle A_3 x - A_3 x', x - x' \rangle \geq L_3^{-1} \|A_3 x - A_3 x'\|^2$ for all x and x' (for example $A_3 = \nabla h$ with h convex and L_3-smooth).

Algorithm 4 Inexact Davis–Yin splitting as an instance of Algorithm 1.

Initialize: $w_0 \in \mathcal{H}$, stepsize $\tau \in \left(0, \frac{2}{L_3}\right)$, relaxation $\theta \in (0, 1 - \frac{\tau L_3}{4}]$, inexactness $\sigma \in [0, 1)$, stride $s \in \mathbb{N}$, and stopping tolerance δ
for $k = 0, 1, 2, \ldots$ **do**
  Produce a pair $(x_k, u_k) \in \mathcal{H} \times \mathcal{H}$ with

$$u_k \in A_1 x_k, \qquad \tau u_k + x_k \approx w_k.$$

  Compute $y_k = J_{\tau A_2}(x_k - \tau u_k - \tau A_3 x_k)$.
  if $\theta \|x_k - y_k\| \le \delta$ **then**
    Stop and return (x_k, y_k, u_k, w_k).
  else if $\|x_k + \tau u_k - w_k\| \le \sigma \|(1-\theta)x_k + \theta y_k + \tau u_k - w_k\|$ **then**
    Set $w_{k+1} = w_k - \theta(x_k - y_k)$.
  else
    Improve the pair (x_k, u_k) with $u_k \in A_1 x_k$ to reduce $\|x_k + \tau u_k - w_k\|$ via the iterative procedure s times and go to line 4.
  end if
end for

3. *B is maximal monotone, and $J = (I + B)^{-1}$ is the relaxed Davis–Yin map*

$$J(w) = w - \theta(x - y), \qquad x = J_{\tau A_1}(w), \quad y = J_{\tau A_2}(2x - w - \tau A_3 x).$$

4. *At termination of Algorithm 4, $u_k \in A_1 x_k$ and $v_k := \frac{1}{\tau}(x_k - y_k) - u_k - A_3 x_k \in A_2 y_k$, with*

$$\|x_k - y_k\| \le \frac{\delta}{\theta}, \qquad \|u_k + v_k + A_3 x_k\| = \frac{1}{\tau}\|x_k - y_k\| \le \frac{\delta}{\tau\theta}.$$

Proof. We prove each item separately.

1. Clearly, the linear map C^* is surjective. Expanding $0 \in A(x_1, x_2, v)$ row by row gives

$$0 \in A(x_1, x_2, v) \iff \begin{cases} v - x_2 \in \tau A_1 x_1, \\ x_1 - v - \tau A_3 x_1 \in \tau A_2 x_2, \\ \theta(x_1 - x_2) = 0. \end{cases}$$

   Adding the first two inclusions and using the fact that $\theta > 0$ gives $-\tau A_3 x \in \tau A_1 x + \tau A_2 x$, which is precisely the desired inclusion. Conversely, if x solves (27), there exists $a_1 \in A_1 x$ with $-a_1 - A_3 x \in A_2 x$. Setting $v = x + \tau a_1$ then solves the inclusion problem.

2. Expanding $C\zeta \in A(x_1, x_2, v)$ row by row,

$$\begin{aligned} \zeta - x_2 + v &\in \tau A_1 x_1, \\ -\zeta - \tau A_3 x_1 + x_1 - v &\in \tau A_2 x_2, \\ \zeta &= \theta(x_1 - x_2). \end{aligned}$$

   Set $\tau u := \zeta - x_2 + v$, so that $u \in A_1 x_1$ and, by the third row, $v = \tau u - \zeta + x_2 = \tau u - \theta x_1 + (1+\theta) x_2$. Substituting into the second row gives $x_1 - \tau u - \tau A_3 x_1 - x_2 \in \tau A_2 x_2$. That is, $x_2 =$

$J_{\tau A_2}(x_1 - \tau u - \tau A_3 x_1)$. Substituting (29) into the three rows shows the converse statement. Then $C^*(x_1, x_2, v) = x_1 - x_2 + v = (1-\theta)x_1 + \theta x_2 + \tau u$, and adding $\zeta = \theta(x_1 - x_2)$ gives $\zeta + C^*(x_1, x_2, v) = x_1 + \tau u$. With $\gamma = 1$ and $(x_1, x_2, u) = (x_k, y_k, u_k)$, the terminal test $\|\zeta\| \le \delta$ of Algorithm 1 is line 5, its relative test is line 7, and its update $w_{k+1} = w_k - \zeta$ is line 8. Line 4 computes the remaining components of the graph pair from (x_k, u_k), so improving (x_k, u_k) and recomputing y_k is the improvement step of Algorithm 1, and the residual $x_k + \tau u_k - w_k$ depends only on the output of the A_1-solver. Since (29) is a function of (x_1, u), starting the A_1-solver at the previously accepted (x_{k-1}, u_{k-1}) reproduces the previously accepted graph pair.

3. Let $(C^*(x_1, x_2, v), \zeta)$ and $(C^*(x_1', x_2', v'), \zeta')$ be two graph pairs, with u, u' as in item 2, and let $a_2 := \frac{1}{\tau}(x_1 - \tau u - \tau A_3 x_1 - x_2) \in A_2 x_2$, and similarly a_2'. Write d_i for the differences of the components, $\bar{d} := d_1 - d_2$, and Δu, Δa_2, $\Delta A_3 := A_3 x_1 - A_3 x_1'$ for the differences of the operator values. From (29), $d_3 = \tau \Delta u - \theta \bar{d} + d_2$ and $\zeta - \zeta' = \theta \bar{d}$, so

$$\left\langle C^*(d_1, d_2, d_3), \zeta - \zeta' \right\rangle = \theta \left\langle \bar{d} + d_3, \bar{d} \right\rangle = \theta(1-\theta)\|\bar{d}\|^2 + \theta\tau \left\langle \Delta u, \bar{d} \right\rangle + \theta \left\langle d_2, \bar{d} \right\rangle.$$

By the definition of a_2, $\bar{d} = \tau \Delta u + \tau \Delta A_3 + \tau \Delta a_2$, hence $\left\langle d_2, \bar{d} \right\rangle = \tau \langle d_2, \Delta u \rangle + \tau \langle d_2, \Delta A_3 \rangle + \tau \langle d_2, \Delta a_2 \rangle$, and therefore

$$\left\langle C^*(d_1, d_2, d_3), \zeta - \zeta' \right\rangle = \theta(1-\theta)\|\bar{d}\|^2 + \theta\tau \left(\langle \Delta u, d_1 \rangle + \langle \Delta a_2, d_2 \rangle + \langle \Delta A_3, d_2 \rangle \right).$$

The first two inner products are nonnegative by monotonicity of A_1 and A_2. By cocoercivity and $d_2 = d_1 - \bar{d}$, $\langle \Delta A_3, d_2 \rangle \ge L_3^{-1} \|\Delta A_3\|^2 - \|\Delta A_3\| \|\bar{d}\|$. Using $1 - \theta \ge \frac{\tau L_3}{4}$,

$$\begin{aligned} \left\langle C^*(d_1, d_2, d_3), \zeta - \zeta' \right\rangle &\ge \theta\tau \left(\frac{L_3}{4} \|\bar{d}\|^2 - \|\Delta A_3\| \|\bar{d}\| + \frac{1}{L_3} \|\Delta A_3\|^2 \right) \\ &= \theta\tau \left(\frac{\sqrt{L_3}}{2} \|\bar{d}\| - \frac{\|\Delta A_3\|}{\sqrt{L_3}} \right)^2 \\ &\ge 0, \end{aligned}$$

so B is monotone. For maximality, fix w and set $x_1 = J_{\tau A_1}(w)$, $\tau u = w - x_1 \in \tau A_1 x_1$. By item 2, this yields a graph pair with $\zeta + C^*(x_1, x_2, v) = w$, so $\mathcal{R}(I + B) = \mathcal{H}$ and B is maximal monotone by Minty's theorem [Bauschke and Combettes, 2017, Theorem 21.1]. The same graph pair gives $J(w) = C^*(x_1, x_2, v) = w - \zeta = w - \theta(x_1 - x_2)$ with $x_2 = J_{\tau A_2}(x_1 - \tau u - \tau A_3 x_1) = J_{\tau A_2}(2x_1 - w - \tau A_3 x_1)$.

4. The definition of y_k in line 4 of Algorithm 4 means $x_k - \tau u_k - \tau A_3 x_k - y_k \in \tau A_2 y_k$, i.e., $v_k \in A_2 y_k$, and then $u_k + v_k + A_3 x_k = \frac{1}{\tau}(x_k - y_k)$. The bounds follow from line 5 of Algorithm 4.

□

Remark 5.1. Item 3 is consistent with the fact that Davis–Yin splitting is $\frac{2}{4 - \tau L_3}$-averaged for $\tau \in (0, \frac{2}{L_3})$ [Davis and Yin, 2017, Proposition 2.1]. Relaxing it by $\theta = 1 - \frac{\tau L_3}{4}$ gives a $\frac{1}{2}$-averaged, i.e., firmly nonexpansive, map, which is the resolvent of a maximal monotone operator. The computation in the proof only uses $0 < \theta \le 1 - \frac{\tau L_3}{4}$, so it shows that B is maximal monotone for every $\tau L_3 < 4$, not only for $\tau < \frac{2}{L_3}$.

Remark 5.2. We also note that the A defined in (28) is not the same as that of Alves et al. [2026, Section 3.3]. Our construction highlights the expressive power of the framework, where the underlying operator need not necessarily be maximal monotone, as long as one can find the appropriate C to make Assumption 2.2 hold.

The ability to represent Davis–Yin splitting is of particular note, since this implies that the framework of Alves et al. [2026] and Bredies et al. [2022] encompasses Douglas–Rachford and its variants, as well as forward-backward splitting. This means that this framework encompasses ADMM, Chambolle–Pock, and proximal gradient descent, among others.

6 Numerical Experiments

We numerically examine the strength of our conclusions and how well the results hold up to empirical scrutiny. We first study the two algorithms listed above: PPM and ADMM, with the goal of examining how inexactness impacts the performance of these algorithms as well as how conservative the theoretical findings are. We also include a study of a rotation operator which highlights more adversarial behavior one could encounter in practice. Here, performance refers both to inner and outer iteration complexity, as well as wall-clock time. All experiments were run on the author's MacBook Pro (Apple M5 Max, 128 GB of memory) in Python 3.14.5 with NumPy 2.5.3, SciPy 1.18.1, CVXPY 1.9.3, and Clarabel 0.11.1. The code is available at https://github.com/stellatogrp/rand_nla_fom. We declare that the code for these experiments was generated with the assistance of Claude, and we take full responsibility for the content of this section.

6.1 Inexact Proximal Point

In this section, we empirically study the algorithm theoretically examined in Proposition 5.1. We test Algorithm 1 on an instance of linear regression with Huber loss [Huber, 1964]. That is,

$$\text{minimize} \quad \textstyle\sum_{i=1}^m h_\rho(a_i^\top x - b_i), \tag{30}$$

where $x \in \mathbb{R}^n$ is the optimization variable, $\{(a_i, b_i)\}_{i=1}^m$ is the data, and $h_\rho : \mathbb{R} \to \mathbb{R}$ is the Huber loss function with parameter ρ:

$$h_\rho(x) = \begin{cases} \frac{1}{2}x^2, & |x| \le \rho, \\ \rho(|x| - \frac{1}{2}\rho), & |x| \ge \rho. \end{cases}$$

We define the objective $f(x) = \sum_{i=1}^m h_\rho(a_i^\top x - b_i)$ to be the sum of the Huber losses of the residuals. The parameter ρ controls the tradeoff between the ℓ_2 and ℓ_1 loss. We set the parameter $\rho = \frac{1}{2}$, and note that the L-smoothness constant of the objective in (30) is $L = \|A\|_2^2$, where A is the design matrix whose rows are a_i. Its gradient can be easily computed in closed form as $\nabla f(x) = A^\top \Pi_{[-\rho,\rho]^m}(Ax - b)$, where $\Pi_{[-\rho,\rho]^m}$ is the projection onto the m-box $[-\rho, \rho]^m$.

We construct synthetic data $\{(a_i, b_i)\}_{i=1}^m$, with $m = 3000$ and $n = 2000$, where $a_i \sim \mathcal{N}(0, m^{-1} I_n)$ are i.i.d. We sample $x_{\text{true}} \sim \mathcal{N}(0, I_n)$ and set $b = Ax_{\text{true}} + 0.1\eta$, where $\eta \sim \mathcal{N}(0, I_m)$. We then select uniformly at random 10% of the entries of b (300 entries) to be perturbed by a Uniform($[-8, -3] \cup [3, 8]$) disturbance. We generate 100 sets of problem data and run each configuration of algorithm parameters on these 100 instances. For a baseline, we compare to the solution time of CVXPY using the Clarabel solver [Agrawal et al., 2018; Diamond and Boyd, 2016; Goulart and Chen, 2026].

The proximal subproblem is to solve

$$\text{prox}_{\gamma f}(x) = \underset{y \in \mathbb{R}^n}{\text{argmin}} \quad \sum_{i=1}^m h_\rho(a_i^\top y - b_i) + \frac{1}{2\gamma}\|y - x\|^2, \tag{31}$$

which has no known closed-form solution. We use (unaccelerated) gradient descent with fixed stepsize $\gamma_{\text{GD}} = \frac{2\gamma}{\gamma L+2}$ to solve the proximal subproblem (31) with proximal parameter $\gamma = \frac{10}{L}$. We run the algorithm until the gradient is sufficiently small: $\|\nabla f(x_k)\| \leq \delta$, and we set $\delta = 10^{-9}$. Since the solution to Huber regression is not known in closed form, this is a practical convergence metric to use. We can substitute this setup into Algorithm 1 directly, letting $C = I$ and $A = \nabla f$. Since this is precisely the scenario of Proposition 5.1, we compare our empirical results against the theoretical guarantees of the algorithm.

Our main focus is to understand the tradeoff between the work required for a given subproblem and the speed of convergence of the outer method. In this spirit, we sweep over the parameters $\sigma \in \{0.01, 0.05, 0.2, 0.5, 0.9, 1\}$ (inexactness) and $s \in \{1, 2, 5, 10, 20\}$ (stride). The σ sweep runs at stride $s = 1$, the stride sweep at $\sigma = 0.1$, and every run starts from $w_0 = 0$. Since the theory guarantees termination only for $\sigma < 1$, every inexact run is also subject to a hard stop: 10^6 inner iterations in total or 300 seconds of wall-clock time, checked at every relative test, in addition to a cap of 50,000 outer iterations. The near-exact baseline solves each subproblem to an inner residual $\gamma\|\nabla\varphi_k\|$ of 10^{-12}. The near-exact baseline always terminates and is subject to the cap on inner iterations only. No run reported here reached any of these limits, including those at $\sigma = 1$.

Figure 1 shows a comparison of the shifted geometric mean (SGM) of the wall-clock times for the different choices of parameters. For this study, choosing the median time or the SGM of the results made no difference in the ordering of configurations. Reducing subproblem accuracy improves the speed of convergence both in wall-clock time and gradient evaluations, as seen in Figures 1 and 2, respectively. In fact, sufficiently inexact iterations outperform running the same gradient descent algorithm on the original problem (30) itself.

Figure 2 compares the convergence behavior of the configurations; Figure 11 in Appendix D.1 magnifies its outer-iteration panels. Despite producing virtually identical suboptimality and stationarity residuals, the performance over the *inner* work is different. In fact, the near-exact PPM performs worse than all other configurations in this regard. Moreover, the highly inexact choices of parameters, $\sigma = 0.5$ and $\sigma = 0.9$, outperform vanilla gradient descent in terms of the number of gradient evaluations. Even $\sigma = 0.2$ is competitive with gradient descent.

The magnified view in Figure 11 confirms that the convergence metrics are nearly indistinguishable, suggesting that there is little effect of inexactness on the outer iterations. The theory predicts a multiplicative factor of $\frac{1+\sigma}{1-\sigma}$ in the worst-case convergence rate, which equals 19 for $\sigma = 0.9$. This factor is not visible in the numerics, highlighting the pessimism of the theory.

The trajectory of the inexact iterates tends to closely follow that of the near-exact algorithm. In Figure 3, the near-exact solutions to the subproblems of the inexact PPM (hollow markers) closely match the trajectory of the near-exact PPM (dashed line). The inexact method required 47 outer iterations ($K_\delta = 47$), but the near-exact method required only 22 ($K_\delta = 22$). However, the inexact PPM with $\sigma = 0.9$ used only 142 gradient evaluations, whereas the near-exact PPM used 1,571 gradient evaluations, over 10 times the number of gradient evaluations.

Additionally, we compare the predicted cost of choosing different values of the stride s (lines 4–10 of Algorithm 1). The relaxed bound refers to that of Lemma 4.2, while the unrelaxed bound refers to Remark 4.4. The unrelaxed bound gives a tighter bound, but only up to a nearly constant factor. Moreover, the conservatism of Lemma 4.2 can be seen in Figure 4, where increasing the stride does not seem to affect the time per outer iteration significantly. The cost of checking depends on factors such as hardware types and availability, difficulty of subproblems, implementation details, and so forth, but using the $s^\star$ defined in Lemma 4.2 seems to be a good heuristic.

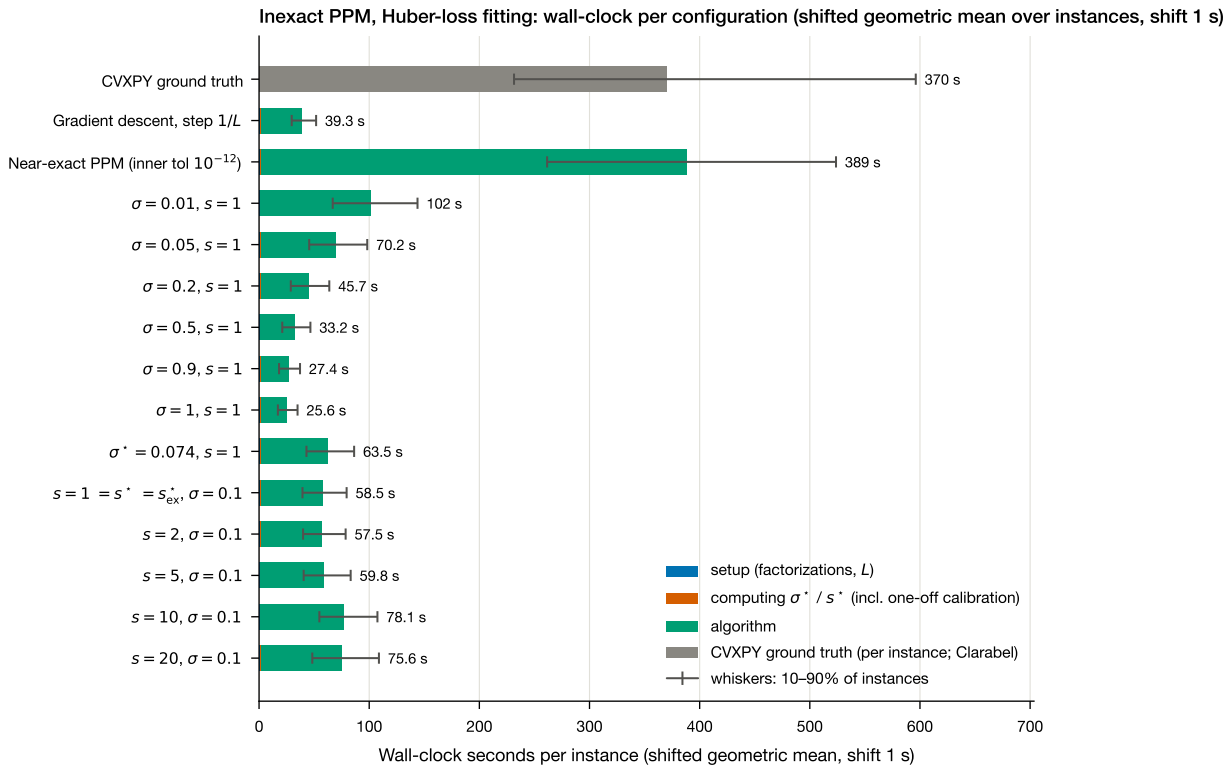


Figure 1: Wall-clock time per configuration on the Huber instances (30) with $n = 2000$ and $m = 3000$: the shifted geometric mean over the 100 instances of the setup time, of the time to compute $\sigma^\star$ or $s^\star$ where one is used (including a one-off calibration), and of the algorithm time, stacked. Whiskers span the 10th to the 90th percentile of the algorithm time, and the top bar is the CVXPY (Clarabel) solve time for scale. The configurations are gradient descent on (30), the near-exact method, the σ sweep at $s = 1$ (including $\sigma^\star$), and the s sweep at $\sigma = 0.1$ (including $s^\star$). Sufficiently inexact runs are faster than both the near-exact method and gradient descent.

6.2 Inexact ADMM

In this section, we empirically study the performance of Algorithm 3 on a quadratic program with linear equality constraints. That is,

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^\top Qx + d^\top x, \\ \text{subject to} & Ax = b, \end{array} \tag{32}$$

where $x \in \mathbb{R}^n$ is the optimization variable and $A \in \mathbb{R}^{m\times n}$, $Q \succ 0$, $d \in \mathbb{R}^n$, and $b \in \mathbb{R}^m$ are the problem data. We take $m = 600$ and $n = 2000$ in these trials, with penalty $c = \frac{L}{10}$, where L is the smoothness constant of the objective. Our stopping tolerance is $\delta = 10^{-9}$. The sweeps are the same as for the Huber instances: the σ sweep runs at $s = 1$ and the stride sweep at $\sigma = 0.1$. Every run starts from $z_0 = p_0 = 0$. As before, we compare to the solution time of CVXPY using the Clarabel solver, but we also include a closed-form solution for the x-subproblem in (33a). Since the optimality conditions for the x-subproblem (33a) are a linear system, $(Q + cI)x = cz_k - d - p_k$, we compare an iterative solution to using a Cholesky factorization of the matrix $Q + cI$. In large-scale applications, obtaining such a factorization may not be feasible, but this is still implemented in some well-known solvers such as OSQP [Stellato et al., 2020].

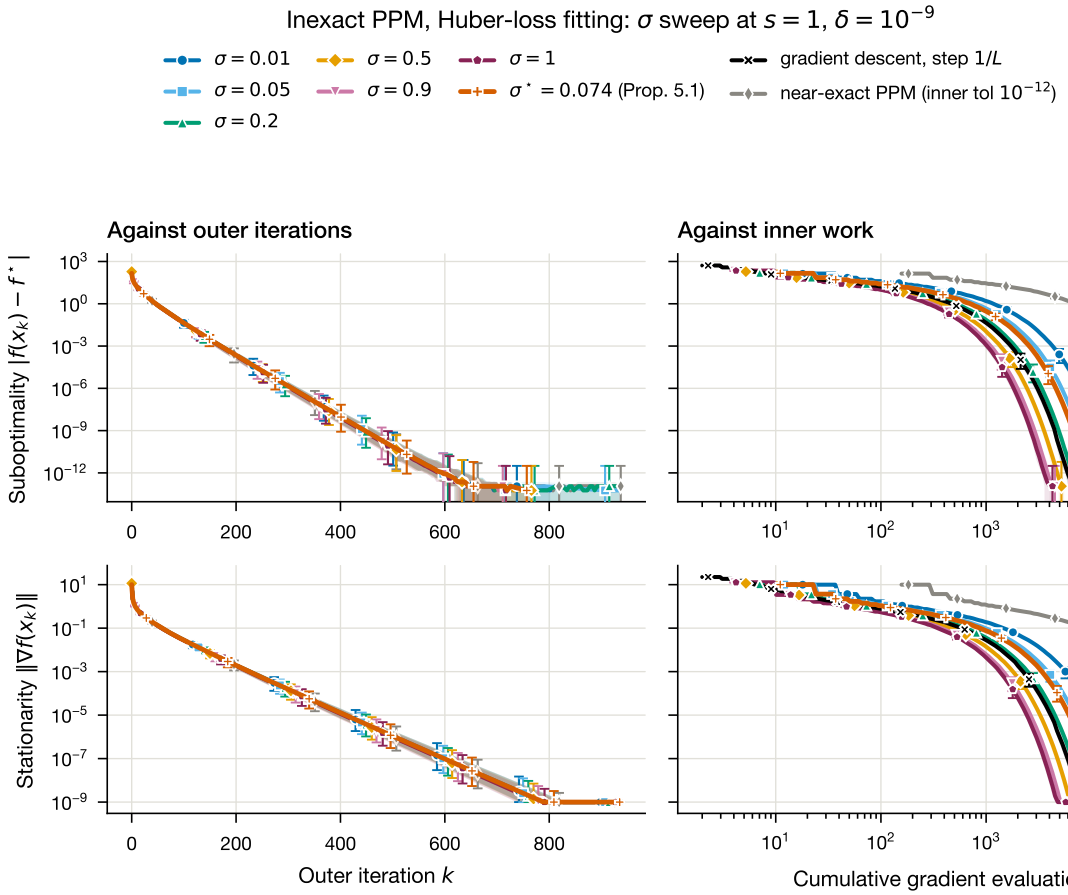


Figure 2: Convergence of Algorithm 1 on the Huber instances (30) for the σ sweep at $s = 1$, for $\sigma^\star$, and for the near-exact method: suboptimality $|f(x_k) - f^\star|$ (top) and stationarity $\|\nabla f(x_k)\|$ (bottom) against outer iterations (left) and against cumulative gradient evaluations (right). Lines are medians over the 100 instances, bands interquartile ranges, and whiskers the 10th to the 90th percentiles; gradient descent has no outer iterations and appears against gradient evaluations only. The curves nearly coincide against outer iterations but separate widely against inner work, where the near-exact method is the slowest.

We sample $Q = V\,\mathbf{diag}(\lambda)V^\top$, where λ is log-spaced in $[\kappa^{-1}, 1]$, V is the orthogonal factor of the QR decomposition of a matrix with i.i.d. $\mathcal{N}(0,1)$ entries, and $\kappa = 10^4$. This also shows that $\|Q\|_2 = 1$ is the L-smoothness constant of the objective. The regularized matrix $Q + cI$ appearing in the x-subproblem (33a) has $\kappa_c := \kappa_2(Q+cI) = \frac{1+c}{10^{-4}+c} \le 1 + \frac{L}{c}$. We use the estimate $\kappa_c \le 1 + \frac{L}{c} = 11$ for the estimation of parameters. In this setup, the exact value is $\kappa_c \approx 10.99$, so this approximation is good for matrices with small eigenvalues. The constraint matrix A has i.i.d. $\mathcal{N}(0, n^{-1})$ entries. We generate ground truth primal and dual solutions $x^\star \in \mathbb{R}^n$ and $\nu^\star \in \mathbb{R}^m$ with i.i.d. $\mathcal{N}(0, n^{-1})$ and $\mathcal{N}(0, m^{-1})$ entries, respectively. The data d and b are computed such that the pair $(x^\star, \nu^\star)$ solves the Karush–Kuhn–Tucker (KKT) system, so that the ADMM dual p_k converges to $-(Qx^\star + d) = A^\top \nu^\star$.

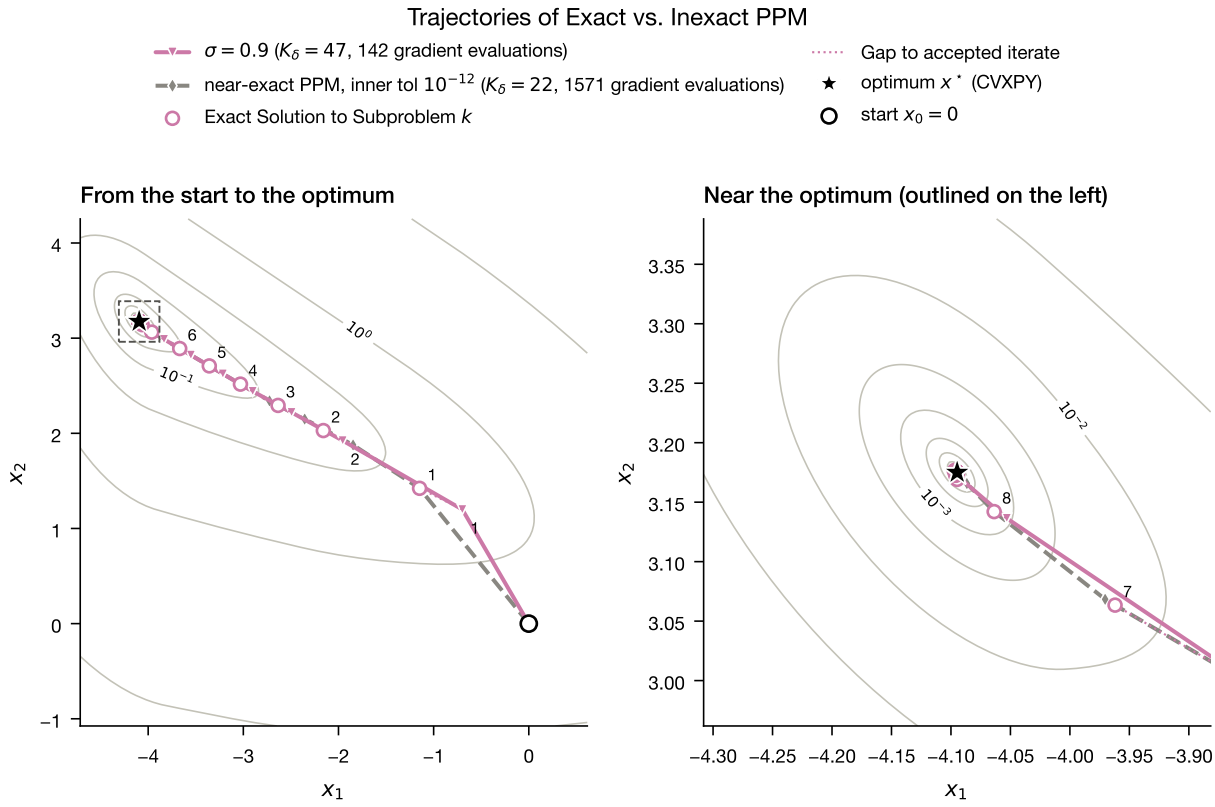


Figure 3: Accepted iterates of the near-exact method (dashed) and of one inexact run with $\sigma = 0.9$ and $s = 1$ on a two-dimensional Huber instance, over contours of $f - f^\star$: left, from the start $x_0 = 0$ to the optimum; right, the neighborhood of the optimum outlined on the left. Hollow markers are the exact solutions $J(w_k)$ of the subproblems that the inexact run approximated, joined to the accepted iterate by dotted segments; a point that passed several consecutive acceptance tests without an inner step carries all their iteration numbers. The inexact run takes 47 outer iterations against 22, but 142 gradient evaluations against 1,571.

Since $Q \succ 0$ and A is full rank, the solution is unique. The subproblems (24) become

$$x_k \approx \operatorname*{argmin}_x \left\{ \frac{1}{2} x^\top Q x + d^\top x + \langle p_k, x \rangle + \frac{c}{2} \|x - z_k\|^2 \right\}, \tag{33a}$$

$$p_{k+1} = p_k + c(x_k - z_k),$$

$$z_{k+1} = \operatorname*{argmin}_z \left\{ \mathcal{I}_{\{z \mid Az = b\}}(z) - \langle p_{k+1}, z \rangle + \frac{c}{2} \|z - x_k\|^2 \right\}. \tag{33b}$$

Substituting this into Algorithm 3 yields the algorithm found in the code. Note that the z-subproblem (33b) is the projection of $x_k + p_{k+1}/c$ (of $x_k - a_k/c$ in Algorithm 3) onto the affine subspace $\{z \mid Az = b\}$, which can be done to machine precision. The inexactness arises from the x-subproblem, which we internally solve using GMRES, since the optimality conditions are a positive definite linear system. We remark that in this setup, $C_0 = 4$ and $\sqrt{\alpha} = \frac{\sqrt{\kappa_c}-1}{\sqrt{\kappa_c}+1}$, so $\alpha \approx 0.288$. The constants come from an application of the Chebyshev bound in the proof of Trefethen and Bau [1997, Theorem 38.5] to Saad and Schultz [1986, Proposition 4].

The wall-clock times in Figure 5 demonstrate that the theory is fairly conservative. Very inexact

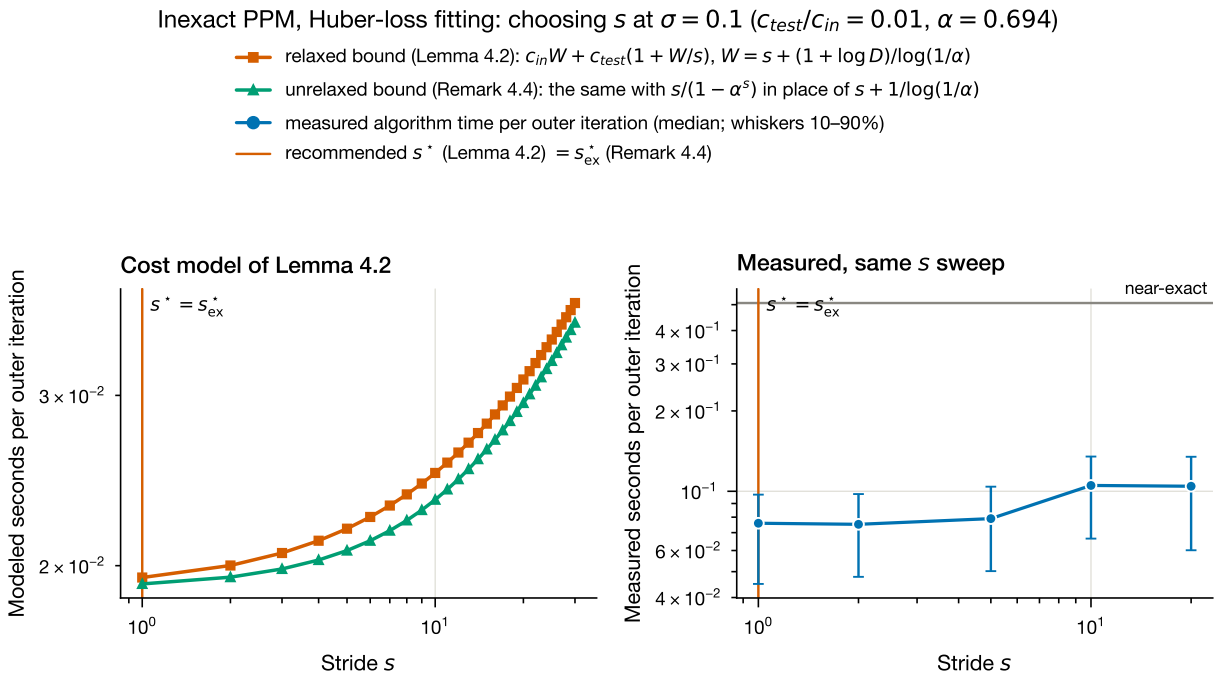


Figure 4: Choosing the stride on the Huber instances. Left, the modeled seconds per outer iteration as a function of s from Lemma 4.2 (relaxed bound) and from Remark 4.4 (unrelaxed bound), with their integer minimizers marked; right, the measured seconds per outer iteration of the s sweep at $\sigma = 0.1$ (median over the 100 instances; whiskers from the 10th to the 90th percentile), with the level of the near-exact method as a horizontal line. The relaxation shifts the model by a nearly constant factor, and the measured time is almost flat in s.

iterations appear to have *better* performance than their exact counterparts. We attribute this to the ability to move in generally the correct direction, even though the precise step may not be accurate. However, all configurations using an iterative method to solve the x-subproblem in (33a) are slower than using the Cholesky factorization of $Q + cI$. We do not consider this to be a limitation, since in many applications such a closed-form solution is either not available or not feasible to implement.

The convergence behavior of the outer iterates in Figure 6 is comparable across all configurations; Figure 12 in Appendix D.2 magnifies its outer-iteration panels. However, solving the subproblem to extremely high absolute accuracy (in this case, 10^{-12}) is unnecessary and often detrimental.

The magnified view in Figure 12 confirms that the convergence metrics are nearly indistinguishable, and in fact the more inexact configurations perform *better* than the exact ones. The inexact configurations outperform the theory, which is generally conservative.

As in the previous section, we compare the modeled cost of different choices of s to the actual measured cost. Numerically solving for a more precise $s^\star$ yields a better result, both theoretically and empirically, as seen in Figure 7. The setup costs of computing $s^\star$ or $\sigma^\star$ are included as the middle segment of each bar, but they are negligible compared to the overall cost of the algorithm and are not visible.

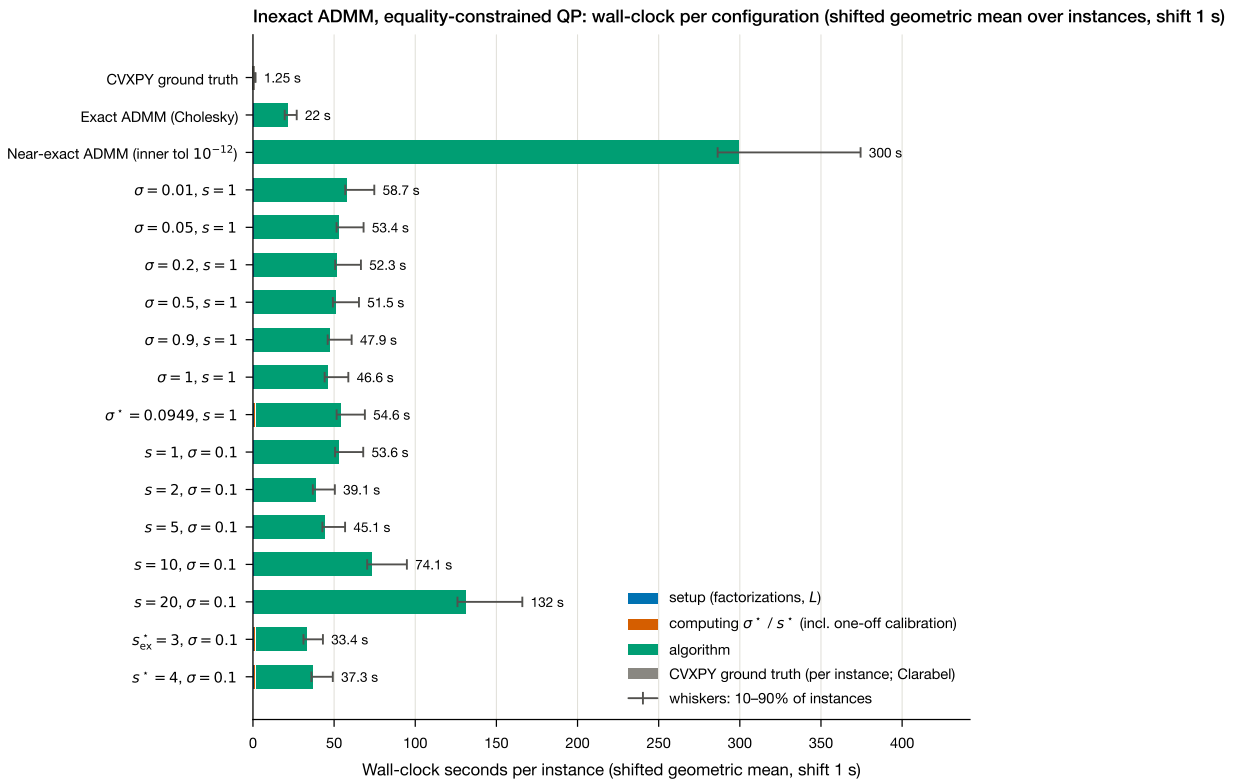


Figure 5: Wall-clock time per configuration on the quadratic programs (32) with $n = 2000$ and $m = 600$: the shifted geometric mean over the 100 instances of the setup time, of the time to compute $\sigma^\star$ or $s^\star$ where one is used (including a one-off calibration), and of the algorithm time, stacked. Whiskers span the 10th to the 90th percentile of the algorithm time, and the top bar is the CVXPY (Clarabel) solve time for scale. The configurations are exact ADMM with a Cholesky factorization of $Q + cI$, the near-exact method, the σ sweep at $s = 1$ (including $\sigma^\star$), and the s sweep at $\sigma = 0.1$ (including $s^\star$). Every configuration that uses GMRES is slower than the factorization, and the inexact ones are faster than the near-exact method.

6.3 Rotation Operator

We first check the one-step decrease of Lemma 3.2 on the two-dimensional Huber instance, and then on an operator for which the lemma is tight for exact steps. On the two-dimensional Huber instance of Figure 3, the decrease of $\|w_k - x^\star\|^2$ at each accepted update is well above the amount that Lemma 3.2 guarantees for every σ (Figure 8). The median over the accepted updates is between 1.77 and 2.58 times $\|\gamma z_k\|^2$ for every σ shown, including $\sigma^\star$, against a guarantee of $\frac{1-\sigma}{1+\sigma}$, so the factor $\frac{1+\sigma}{1-\sigma}$ of Theorem 3.1 is not reflected in the decrease per accepted step. To show that this factor is not an artifact of the analysis, we repeat the experiment in two dimensions on the linear operator $B(x) = U(x - x^\star)$, where $U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is the rotation by a quarter turn. The operator B is monotone (but not a gradient). The instance has $x^\star$ at distance 5 from the start $x_0 = 0$, and the runs use the same σ sweep as before at $s = 1$, with $\delta = 10^{-9}$ and an inner residual of 10^{-12} for the

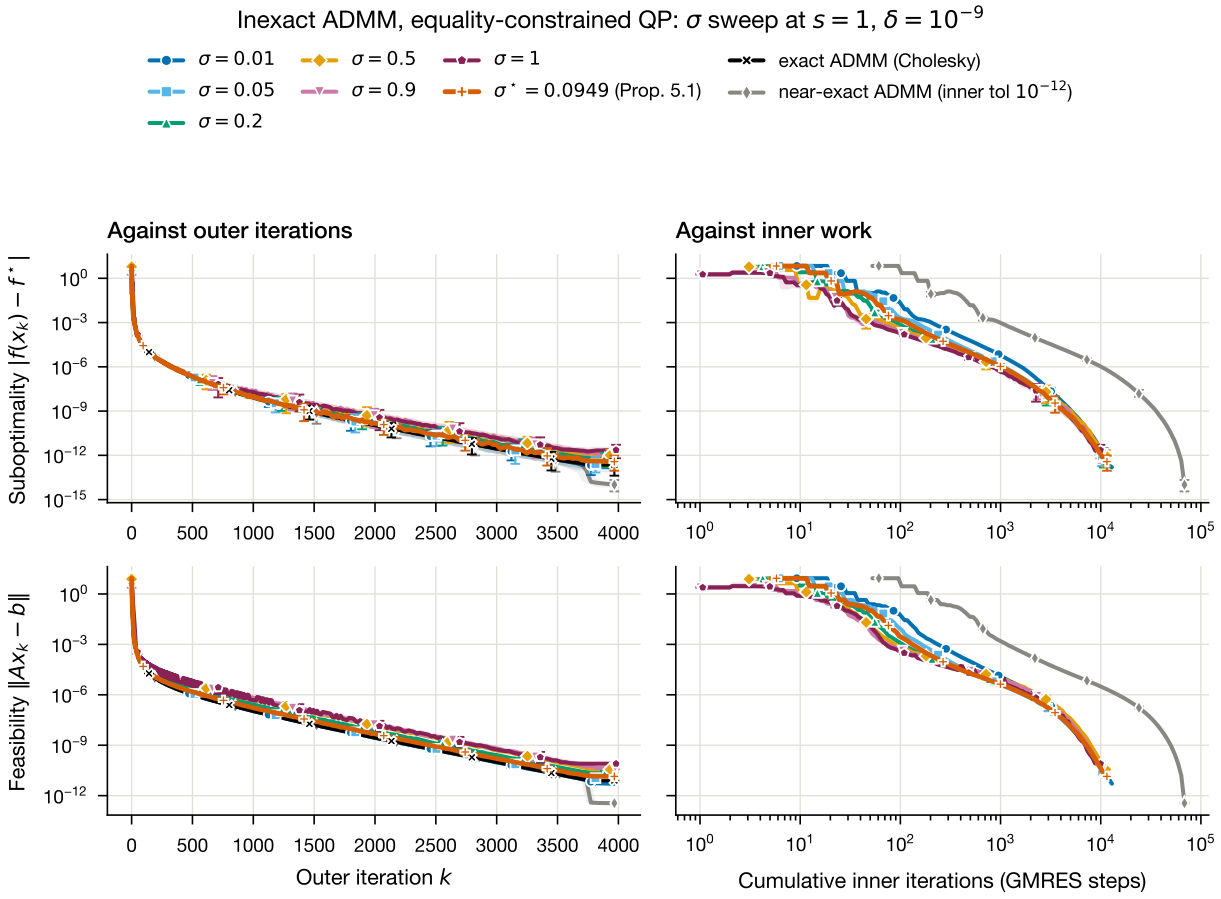


Figure 6: Convergence of Algorithm 3 on the quadratic programs (32) for the σ sweep at $s = 1$, for $\sigma^\star$, for the near-exact method, and for exact ADMM: suboptimality $|f(x_k) - f^\star|$ (top) and primal feasibility $\|Ax_k - b\|$ (bottom) against outer iterations (left) and against cumulative GMRES iterations (right). Lines are medians over the 100 instances, bands interquartile ranges, and whiskers the 10th to the 90th percentiles; exact ADMM has no inner iterations and appears against outer iterations only. The outer-iteration curves are comparable across configurations, while the more inexact ones need the fewest GMRES iterations.

near-exact method. The exact proximal step w^+ satisfies $\|w - x^\star\|^2 - \|w^+ - x^\star\|^2 = \|w - w^+\|^2$[5] and Lemma 3.2 is tight for exact steps. The inner solver is Richardson iteration on the resolvent residual with the step that minimizes its contraction factor, which satisfies Assumption 4.1 with $C_0 = 1$ and $\alpha = \frac{\gamma^2}{1+\gamma^2} = 0.8$ for $\gamma = 2$. Figure 9 shows the paths and Figure 10 the observed decrease against its guarantee; the convergence of the sweep, in the layout of Figure 2, is Figure 13 in Appendix D.3. The near-exact method needs $K_\delta = 26$ outer iterations, $\sigma = 0.5$ needs 62, $\sigma = 0.9$ needs 48, and $\sigma = 1$ needs 235, and the median decrease per $\|\gamma z_k\|^2$ is 1.00 for the near-exact method, 1.02 at $\sigma = 0.01$, 0.41 at $\sigma = 0.5$, 0.28 at $\sigma = 0.9$, and 0.047 at $\sigma = 1$, against guarantees of 1, 0.98, 0.33, 0.053, and 0.

[5]The critical property here is skew-symmetry of U. One could take any higher-dimensional skew-symmetric matrix to get the same exact decrease equality.

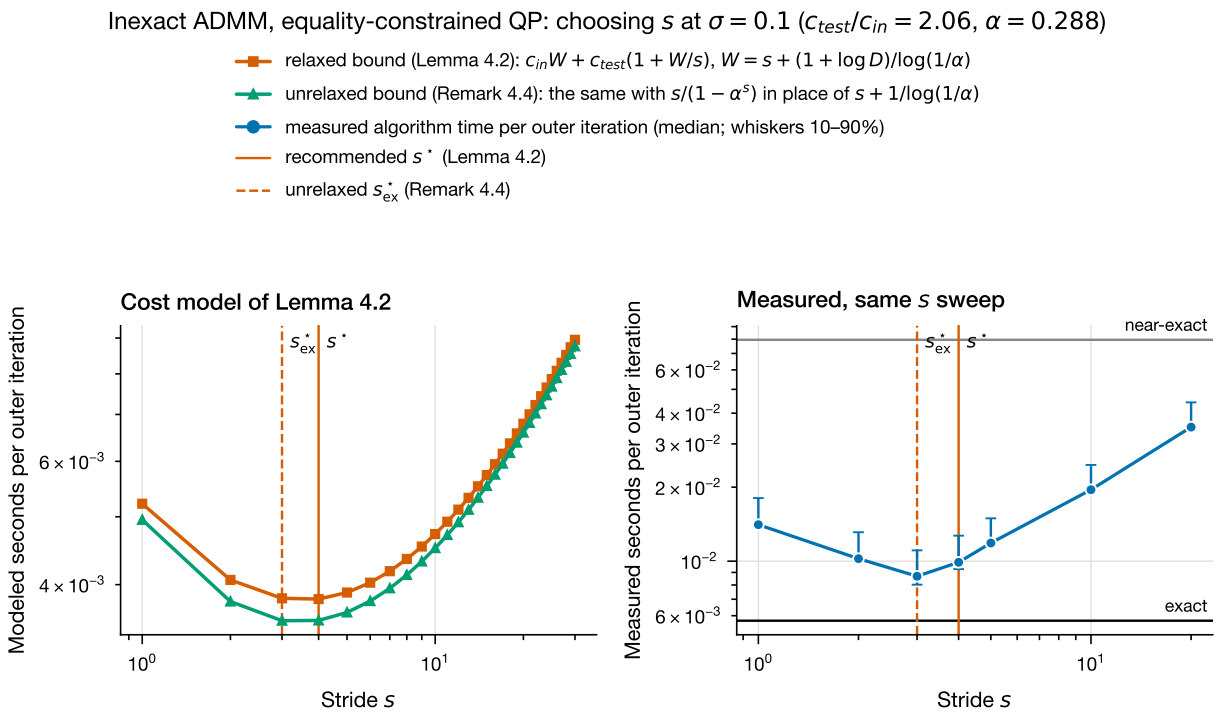


Figure 7: Choosing the stride on the quadratic programs. Left, the modeled seconds per outer iteration as a function of s from Lemma 4.2 (relaxed bound) and from Remark 4.4 (unrelaxed bound), with their integer minimizers marked; right, the measured seconds per outer iteration of the s sweep at $\sigma = 0.1$ (median over the 100 instances; whiskers from the 10th to the 90th percentile), with the levels of exact ADMM and of the near-exact method as horizontal lines. The minimizer of the unrelaxed bound is the better choice, both in the model and in the measurement.

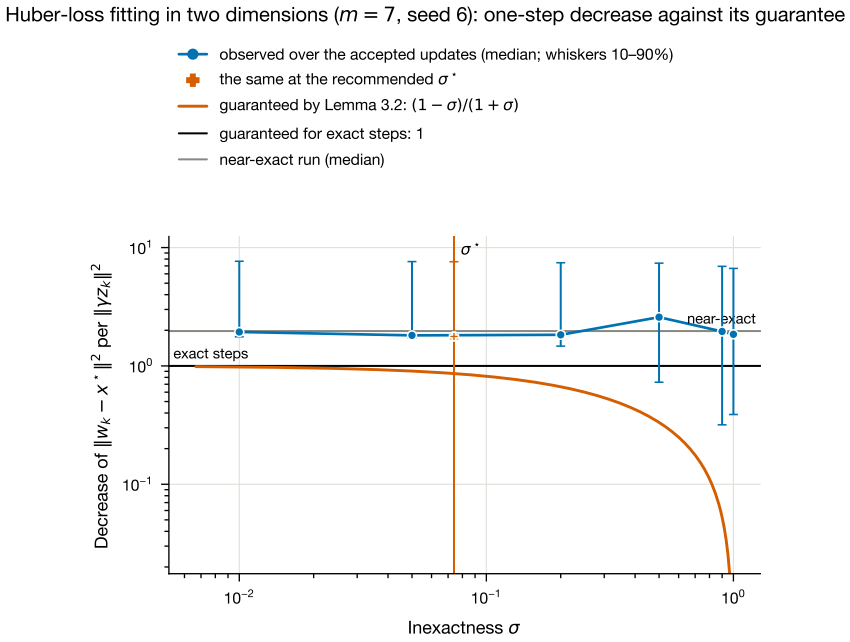


Figure 8: The decrease of $\|w_k - x^\star\|^2$ at each accepted update divided by $\|\gamma z_k\|^2$ on the two-dimensional Huber instance of Figure 3, for the σ sweep at $s = 1$ and for $\sigma^\star$ (median over the accepted updates; whiskers from the 10th to the 90th percentile), against the guarantee $\frac{1-\sigma}{1+\sigma}$ of Lemma 3.2, the level 1 guaranteed for exact steps, and the level observed for the near-exact method. The observed decrease stays near the level of exact steps for every σ.

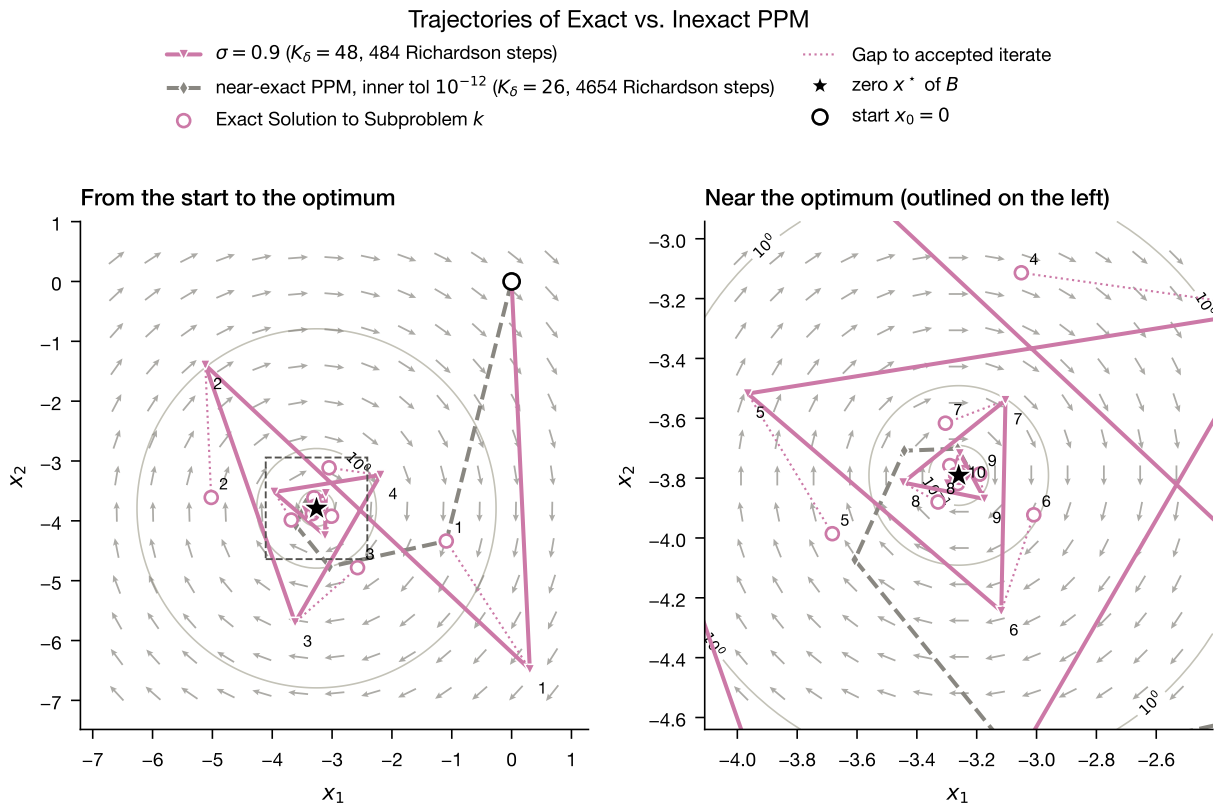


Figure 9: Accepted iterates of Algorithm 1 on the rotation operator $B(x) = U(x - x^\star)$ for $\sigma = 0.9$ (solid) and for the near-exact method (dashed), at $s = 1$, over contours of $\|x - x^\star\|$; arrows give the direction of $-B$, and the right panel magnifies the neighborhood of $x^\star$ outlined on the left. Since B is skew-symmetric, each exact step decreases $\|w - x^\star\|^2$ by exactly $\|w - w^+\|^2$, while the inexact iterates wander around $x^\star$ and take 48 outer iterations against 26.

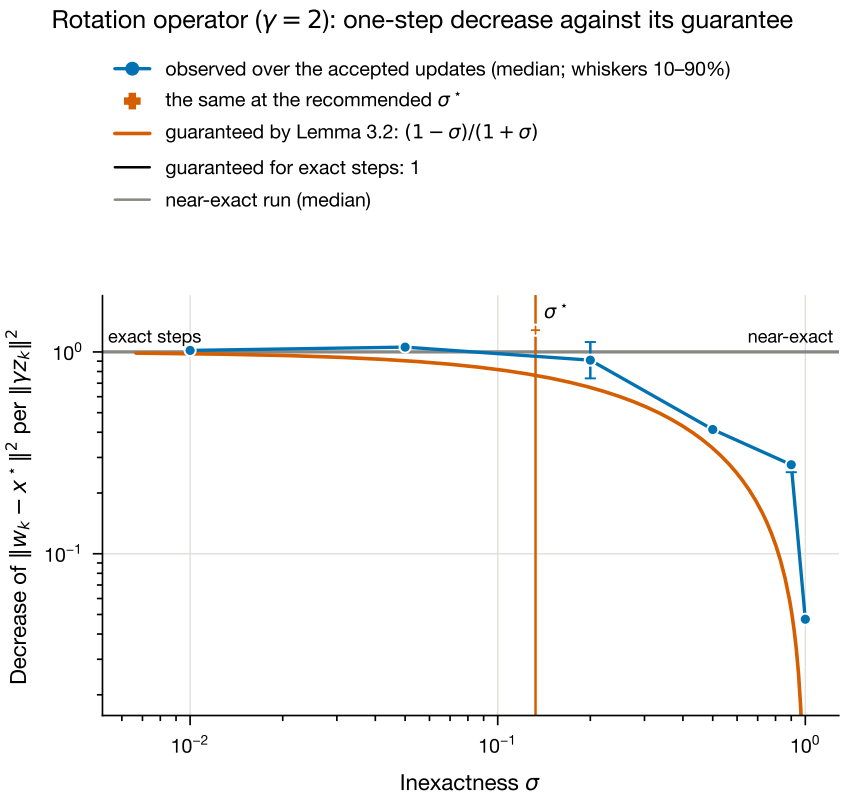


Figure 10: The decrease of $\|w_k - x^\star\|^2$ at each accepted update divided by $\|\gamma z_k\|^2$ on the rotation operator of Figure 9, for the σ sweep at $s = 1$ and for $\sigma^\star$ (median over the accepted updates; whiskers from the 10th to the 90th percentile), against the guarantee $\frac{1-\sigma}{1+\sigma}$ of Lemma 3.2, the level 1 guaranteed for exact steps, and the level observed for the near-exact method. The near-exact run attains the exact level, and the inexact runs stay above their guarantee but drop well below the exact level for $\sigma \geq 0.5$.

7 Conclusion

We analyzed the expected inner work of proximal point methods for solving a monotone inclusion problem in a Hilbert space. Equipped with a contractive inner procedure for computing the resolvent, we provide a quantitative bound on the amount of expected inner work required to solve the monotone inclusion problem to a given accuracy. Our framework encompasses a broad variety of popular operator splitting algorithms, and we provide principled heuristics for selecting algorithm parameters a priori. Numerical experiments validate our findings and suggest that in practice, many algorithms can tolerate significant amounts of inexactness.

Many questions remain open. The $\mathbf{E}[K_\delta]$ term in Theorem 4.2 opens the door to considering average-case complexity, rather than the worst-case performance. Moreover, the choice of problem parameters is not only problem-sensitive, but also sensitive to the choice of inner procedure used. Providing more thorough guidelines on the selection of parameters would be useful in practice. Also of interest is differentiating which inner solvers are more compatible with certain outer solvers, and providing guidance on when to use a particular splitting or method over another. Another promising direction is an adaptive choice of parameters. The analysis of this work assumes that algorithm parameters are fixed, but in practice this is undesirable. Incorporating an analysis of adaptive parameter selection would bring the theory closer to practice. Another direction would be to relax the requirement in Algorithm 1 that $Cz_k \in Au_k$, for cases where finding a member of Au_k is difficult, such as the case where the proximal step is projection onto a convex set.

A Tightness of Bounds

Here, we consider the tightness of the bounds of Section 4.

A.1 Dependence on the Stopping Tolerance

The additive logarithm in Theorem 4.2 and the worst-case rate in Theorem 3.1 describe different regimes. The following two constructions establish their respective dependence on δ.

Proposition A.1 (Necessity of the additive logarithm)**.** *Fix $\sigma \in (0,1)$, $\alpha \in (0,1)$, and a positive integer s. There is a scalar strongly convex quadratic, with $\gamma = 1$, a fixed initialization, and gradient descent as the inner procedure, satisfying Assumption 4.1 with $C_0 = 1$, for which every $\delta \in (0,1)$ gives*

$$K_\delta = 1, \qquad \sum_{k=0}^{K_\delta} N_k = s \left\lceil \frac{\log(1/\delta^2)}{s \log(1/\alpha)} \right\rceil. \tag{34}$$

The right-hand side of (12) in Theorem 4.2 exceeds this work by an additive term not dependent on δ.

Proof. Let $\mathcal{H} = \mathcal{D} = \mathbb{R}$, $C = I$, $\gamma = 1$, and $A = B = \nabla f$, where

$$f(y) = \frac{\mu}{2} y^2, \qquad \mu = 1 + \frac{1}{\sigma}.$$

Set $w_0 = 1$ and initialize the first graph pair at $y^0 = \frac{1}{\mu}$ and $z^0 = 1$. At every outer iteration, apply gradient descent to $f(y) + \frac{1}{2}(y - w_k)^2$ with stepsize $(1 - \sqrt{\alpha})/(1 + \mu)$, using the graph pair $z^l = \mu y^l$.

By Nesterov [2013, Theorem 2.1.15], the update and residual satisfy

$$y^{l+1} = y^l - \frac{1-\sqrt{\alpha}}{1+\mu}\left((1+\mu)y^l - w_k\right),$$
$$e^{l+1} = \sqrt{\alpha}\, e^l, \qquad e^l = (1+\mu)y^l - w_k.$$

Hence the deterministic contraction holds with equality, with the same α and $C_0 = 1$ at every outer iteration.

At the first check, the terminal test fails because $z^0 = 1 > \delta$, whereas the relative test holds with equality:

$$|z^0 + y^0 - w_0| = \frac{1}{\mu} = \sigma\left(1 - \frac{1}{\mu}\right) = \sigma|y^0 - w_0|.$$

Thus $N_0 = 0$, the accepted pair is $(y_0, z_0) = (\frac{1}{\mu}, 1)$, and $w_1 = w_0 - z_0 = 0$. Warm-starting at this pair gives, during the second inner solve,

$$y^l = \frac{\alpha^{l/2}}{\mu}, \qquad z^l = \alpha^{l/2}, \qquad |e^l| = \left(1 + \frac{1}{\mu}\right)\alpha^{l/2} > \frac{\sigma}{\mu}\alpha^{l/2} = \sigma|y^l - w_1|.$$

The relative test therefore fails at every finite inner iteration. The terminal test first passes at the first multiple l of s with $\alpha^{l/2} \le \delta$, proving (34).

For this choice of initialization, $R(w_0) = \frac{\mu}{1+\mu} = \frac{\sigma+1}{2\sigma+1}$ and

$$\frac{(1+\sigma)^2}{\sigma^2}|z^0 + y^0 - w_0|^2 = 1.$$

Consequently, the two logarithmic initialization terms in (12) have

$$\begin{aligned}
(\log\beta_0)_+ + \log\frac{\max\left\{\|R(w_0)\|^2, \gamma^2\delta^2\right\}}{\gamma^2\delta^2} &= \left(\log\frac{1}{\max\left\{\left(\frac{\mu}{1+\mu}\right)^2, \delta^2\right\}}\right)_+ + \log\frac{\max\left\{\left(\frac{\mu}{1+\mu}\right)^2, \delta^2\right\}}{\delta^2} \\
&= \left(\log\frac{1}{\max\left\{\left(\frac{\sigma+1}{2\sigma+1}\right)^2, \delta^2\right\}}\right)_+ + \log\frac{\max\left\{\left(\frac{\sigma+1}{2\sigma+1}\right)^2, \delta^2\right\}}{\delta^2} \\
&= \log(\delta^{-2}),
\end{aligned}$$

where we used the fact that $\max\left\{\left(\frac{\sigma+1}{2\sigma+1}\right)^2, \delta^2\right\} < 1$. Since $K_\delta = 1$, the right-hand side of the total inner work bound (12) of Theorem 4.2 is

$$\frac{2s}{1-\alpha^s} + \frac{\log D_\sigma}{\log(1/\alpha)} + \frac{\log(\delta^{-2})}{\log(1/\alpha)}.$$

Noting that the ceiling in (34) adds fewer than s iterations to the last term completes the proof. □

Remark A.1 (Any prescribed positive number of outer updates). The construction of Proposition A.1 extends to any prescribed positive integer m if the gradient descent stepsize may depend on the outer state. Fix $\sigma \in (0,1)$, $\alpha \in (0,1)$, and a positive integer s, and choose

$$\mu \geq \max\left\{1 + \frac{1}{\sigma}, \alpha^{-s/4}\right\}, \qquad q := \sqrt{\alpha}, \qquad \rho := \mu^{-2/s} \leq q.$$

Use the same quadratic $f(y) = \frac{\mu}{2}y^2$, with $C = I$ and $\gamma = 1$, for every m. At outer state w, apply gradient descent to $f(y) + \frac{1}{2}(y - w)^2$ with the stepsize

$$\eta(w) = \frac{1}{1+\mu}\begin{cases} 1, & w > 1, \\ 1 - \rho, & w = 1, \\ 1 - q, & w < 1. \end{cases}$$

This rule is independent of m and δ. Every graph residual contracts in one inner iteration by a factor of 0, ρ, or q, respectively, so Assumption 4.1 holds with the prescribed α and $C_0 = 1$.

For $m \geq 2$, set $w_0 = (1+\mu)^{m-1}$ and initialize the first graph pair at $y^0 = \frac{w_0}{1+\mu}$ and $z^0 = \mu y^0$. For $m = 1$, set $w_0 = 1$, $y^0 = \frac{1}{\mu}$, and $z^0 = 1$. For every $\delta \in (0,1)$, this gives

$$K_\delta = m, \qquad \sum_{k=0}^{K_\delta} N_k = s(m-1) + s\left\lceil \frac{\log(1/\delta^2)}{s\log(1/\alpha)} \right\rceil. \tag{35}$$

To verify the claim for $m \geq 2$, note that the first $m-1$ accepted updates are exact proximal steps, with

$$w_k = (1+\mu)^{m-1-k} \quad (0 \leq k \leq m-1).$$

The initial pair passes the relative test without inner work. At each subsequent warm start before $w = 1$, the previous exact pair has $y^0 = w_k$ and $z^0 = \mu w_k > \delta$, so both tests fail. The inner solver reaches the exact pair after one iteration, which is checked after s iterations. At $w_{m-1} = 1$, the warm start is $(y^0, z^0) = (1, \mu)$, with residual $e^0 = \mu$. After s inner iterations, $e^s = \rho^s \mu = \frac{1}{\mu}$, giving the graph pair $(y^s, z^s) = (\frac{1}{\mu}, 1)$. This pair passes the relative test because $\frac{1}{\mu} \leq \sigma(1 - \frac{1}{\mu})$, and its accepted update gives $w_m = 0$. All preceding accepted operator values are at least 1, so none passes the terminal test for $\delta < 1$. The final warm-started inner solve has $z^l = q^l$ and never passes the relative test, exactly as in Proposition A.1. This proves (35), including $m = 1$ with its specified initialization.

The actual work and the upper bound in (12) are both $\Theta(m + \log(1/\delta))$, with constants depending only on the fixed parameters. Indeed, for $m \geq 2$ the cold start is exact, so $\beta_0 = 0$, while $R(w_0) = \mu(1+\mu)^{m-2}$. The endpoint logarithm is therefore $2\log\mu + 2(m-2)\log(1+\mu) + \log(1/\delta^2)$. In this family, the initial distance $(1+\mu)^{m-1}$ grows with m. Alternatively, rescaling every iterate and the stepsize rule's threshold by $h_m = (1+\mu)^{-(m-1)}$ gives initial distance 1, with $K_\delta = m$ for $0 < \delta < h_m$ and total work

$$s(m-1) + s\left\lceil \frac{\log(h_m^2/\delta^2)}{s\log(1/\alpha)} \right\rceil.$$

Proposition A.2 (Matching worst-case accuracy bounds)**.** *Fix $\sigma \in (0,1)$ and $\alpha \in (0,1)$, and set $\gamma = s = C_0 = 1$. For every $\delta \in (0, 1/2)$, there is an instance on $\mathbb{R}^2$ with* $\mathbf{dist}(w_0, \mathbf{zer}\, B) = 1$ *and an*

inner procedure satisfying Assumption 4.1 for which

$$\sum_{k=0}^{K_\delta} N_k = K_\delta + 1 = \left\lceil \frac{\log 4}{\log(1+4\delta^2)} \right\rceil \geq \frac{\log 4}{4\delta^2}. \tag{36}$$

The upper bound in (12) is $O(\delta^{-2})$ on this family, with a constant independent of δ.

Proof. Let $\mathcal{H} = \mathcal{D} = \mathbb{R}^2$, $C = I$, $w_0 = (1, 0)$, and

$$A = B = 2\delta U, \qquad U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The operator is monotone because U is skew-symmetric. Since $I + 2\delta U$ is invertible, Minty's Theorem shows that U is maximal monotone [Bauschke and Combettes, 2017, Theorem 21.1]. Its unique zero is the origin, and

$$J = \frac{I - 2\delta U}{1 + 4\delta^2}, \qquad J^\top J = \frac{I}{1 + 4\delta^2}.$$

Initialize the first graph pair at $y^0 = w_0$ and $z^0 = 2\delta U w_0$. Choose the inner procedure that returns $(J(w_k), 2\delta U J(w_k))$ after one inner iteration and retains this exact pair at all subsequent inner iterations. This procedure satisfies the deterministic contraction with $C_0 = 1$ and any prescribed $\alpha \in (0, 1)$, since its residual vanishes after the first iteration. The displayed formula for J evaluates this pair with a fixed number of scalar arithmetic operations.

At the initial check, $y^0 = w_0$ and $\|z^0\| = 2\delta > \delta$, so both tests fail. At every subsequent warm start, $y^0 = y_{k-1} = w_k$, while $z^0 = z_{k-1}$ has norm greater than δ because the preceding update was nonterminal. Thus both tests again fail before the first inner iteration, and $N_k = 1$ for every $0 \leq k \leq K_\delta$. All accepted updates are exact proximal steps, so

$$w_k = J^k w_0, \qquad y_k = J^{k+1} w_0, \qquad \|z_k\| = \frac{t}{(1+4\delta^2)^{(k+1)/2}}.$$

The first terminal check is therefore characterized by

$$K_\delta + 1 = \left\lceil \frac{\log(\frac{4\delta^2}{\delta^2})}{\log(1+4\delta^2)} \right\rceil = \left\lceil \frac{\log 4}{\log(1+4\delta^2)} \right\rceil.$$

The inequality $\log(1 + x) \leq x$ proves the lower bound in (36). Conversely, $\log(1 + x) \geq \frac{x}{1+x}$ gives

$$K_\delta + 1 \leq \frac{\log 4}{4\delta^2} + \log 4 + 1.$$

Moreover,

$$\frac{\|R(w_0)\|^2}{\delta^2} = \frac{4}{1 + 4\delta^2}, \qquad \beta_0 = \frac{(1+\sigma)^2}{\sigma^2}(1 + 4\delta^2).$$

Both quantities are bounded independently of δ, so (12) is also $O(\delta^{-2})$ on this family. □

Remark A.2 (Two subspaces). The resolvent in Proposition A.2 is also a Douglas–Rachford map for two lines. Let P_1 and P_2 be the orthogonal projectors onto $\{(a, 0) \mid a \in \mathbb{R}\}$ and $\{(a, -2\delta a) \mid a \in \mathbb{R}\}$, respectively. Direct multiplication gives

$$I - P_1 + P_2(2P_1 - I) = \frac{1}{1 + 4\delta^2} \begin{bmatrix} 1 & 2\delta \\ -2\delta & 1 \end{bmatrix} = J.$$

Thus the same trajectory and stopping residual occur for exact Douglas–Rachford splitting between these subspaces.

B Proof of Lemma 4.1

B.1 Proof of Bounds G_σ and H_σ

Proof. From the update rule of Algorithm 1,

$$\begin{aligned}\gamma z_k + C^* u_k - w_{k+1} &= \gamma z_k + y_k - (w_k - \gamma z_k) \\ &= e_k + \gamma z_k.\end{aligned}$$

Write $r := \|R(w_k)\|$, $e_J := y_k - J(w_k)$, $e_R := \gamma z_k - R(w_k)$, and $\varepsilon := \|e_k\|$, so that $e_k = e_J + e_R$, $\gamma z_k = R(w_k) + e_R$, $y_k - w_k = e_J - R(w_k)$, and

$$\begin{aligned}\gamma z_k + C^* u_k - w_{k+1} &= e_k + \gamma z_k \\ &= R(w_k) + e_J + 2e_R.\end{aligned}$$

We use three facts. First, (4) in Lemma 3.1 gives $\|e_J\|^2 + \|e_R\|^2 \le \varepsilon^2$. In particular, $\langle e_J, e_R\rangle \ge 0$, so $\|e_R\| \le \varepsilon$ and $\|e_R - e_J\|^2 = \varepsilon^2 - 4\langle e_J, e_R\rangle \le \varepsilon^2$. Second, the relative test gives $\|R(w_k) - e_J\| = \|y_k - w_k\| \ge \frac{\varepsilon}{\sigma}$. Third, by the bound $\|\gamma z + y - w\| \le \frac{\sigma}{1-\sigma}\|R(w)\|$ (7) in Lemma 3.1, $\varepsilon \le \frac{\sigma r}{1-\sigma}$.

If $\varepsilon \le \sigma r$, then

$$\|R(w_k) + e_J + 2e_R\| \le r + \|e_J + e_R\| + \|e_R\| \le r + 2\varepsilon \le (1 + 2\sigma)r \le G_\sigma r.$$

Suppose now $\varepsilon \ge \sigma r$. Expanding, and using $\|R(w_k) + e_J\|^2 = 2r^2 + 2\|e_J\|^2 - \|R(w_k) - e_J\|^2$ and $4\langle e_J, e_R\rangle + 4\|e_R\|^2 = 4\varepsilon^2 - 4\|e_J\|^2 - 4\langle e_J, e_R\rangle$,

$$\begin{aligned}\|R(w_k) + e_J + 2e_R\|^2 &= \|R(w_k) + e_J\|^2 + 4\langle R(w_k), e_R\rangle + 4\langle e_J, e_R\rangle + 4\|e_R\|^2 \\ &= 2r^2 - 2\|e_J\|^2 - \|R(w_k) - e_J\|^2 + 4\varepsilon^2 + 4\langle R(w_k), e_R\rangle - 4\langle e_J, e_R\rangle.\end{aligned}$$

Since $\langle e_J, e_R\rangle \ge 0$, we have $\|R(w_k) - e_J\|^2 \ge \frac{\varepsilon^2}{\sigma^2}$, and

$$\langle R(w_k), e_R\rangle = \langle R(w_k), e_J\rangle + \langle R(w_k), e_R - e_J\rangle \le \langle R(w_k), e_J\rangle + r\varepsilon.$$

Moreover,

$$2\langle R(w_k), e_J\rangle = r^2 + \|e_J\|^2 - \|R(w_k) - e_J\|^2 \le r^2 + \|e_J\|^2 - \frac{\varepsilon^2}{\sigma^2}.$$

Therefore

$$\|R(w_k) + e_J + 2e_R\|^2 \le 4r^2 + 4r\varepsilon + \left(4 - \frac{3}{\sigma^2}\right)\varepsilon^2 =: \Phi(\varepsilon), \tag{37}$$

a quadratic in ε alone. Maximizing Φ over $\varepsilon \in [\sigma r, \frac{\sigma r}{1-\sigma}]$ yields (11).

We now move on to H_σ. Note that $w_{k+1} = w_k - \gamma z_k = J(w_k) - e_R$. Since R is firmly nonexpansive (Lemma 3.1) and $w_{k+1} - w_k = -\gamma z_k = -(R(w_k) + e_R)$,

$$\|R(w_{k+1}) - R(w_k)\|^2 \le \langle R(w_{k+1}) - R(w_k), w_{k+1} - w_k\rangle = -\langle R(w_{k+1}) - R(w_k), R(w_k) + e_R\rangle.$$

As a consequence,

$$\|R(w_{k+1})\|^2 \le \langle R(w_{k+1}), R(w_k) - e_R\rangle + \langle R(w_k), e_R\rangle.$$

The fact that $\langle R(w_{k+1}), R(w_k) - e_R\rangle \le \frac{1}{2}\|R(w_{k+1})\|^2 + \frac{1}{2}\|R(w_k) - e_R\|^2$ and using $\frac{1}{2}\|R(w_k) - e_R\|^2 + \langle R(w_k), e_R\rangle = \frac{1}{2}\left(r^2 + \|e_R\|^2\right)$ gives

$$\|R(w_{k+1})\|^2 \le r^2 + \|e_R\|^2 = \|R(w_k)\|^2 + \|\gamma z_k - R(w_k)\|^2. \tag{38}$$

The first two facts give $\|e_R\|^2 \le \varepsilon^2 - \|e_J\|^2 \le \sigma^2\|R(w_k) - e_J\|^2 - \|e_J\|^2 \le \sigma^2(\|e_J\| + r)^2 - \|e_J\|^2$, a concave quadratic in $\|e_J\|$ whose maximum, attained at $\|e_J\| = \frac{\sigma^2 r}{1-\sigma^2}$, equals $\frac{\sigma^2 r^2}{1-\sigma^2}$. Hence $\|e_R\| \le \frac{\sigma r}{\sqrt{1-\sigma^2}}$ and $\|R(w_{k+1})\|^2 \le r^2 + \frac{\sigma^2 r^2}{1-\sigma^2} = \frac{r^2}{1-\sigma^2} = H_\sigma^2 r^2$.

The proof of the tightness of the constants can be found in Appendix B.2. □

B.2 Tightness of G_σ and H_σ

Proof. The constant G_σ in (10a) is sharp. Fix σ and $r > 0$, let $\varepsilon \in [\sigma r, \frac{\sigma r}{1-\sigma}]$ maximize Φ in (37), and set, for an orthonormal pair (v_1, v_2),

$$\begin{aligned} e_J &= \|e_J\| v_1, \\ e_R &= \|e_R\| v_2, \\ \gamma\bar{z} &= \frac{r}{\varepsilon}(-\|e_J\| v_1 + \|e_R\| v_2), \end{aligned}$$

which is possible because $\|e_J\|^2 \ge 0$ for $\varepsilon \ge \sigma r$ and $\|e_R\|^2 \ge 0$ for $\varepsilon \le \frac{\sigma r}{1-\sigma}$. This definition is not circular since we have

$$\begin{aligned} \|e_J\|^2 &= \frac{\varepsilon(\varepsilon^2 - \sigma^2 r^2)}{\sigma^2(\varepsilon + 2r)}, \\ \|e_R\|^2 &= \varepsilon^2 - \|e_J\|^2. \end{aligned}$$

Moreover, $\|\gamma\bar{z}\| = r$, $\langle e_J, e_R\rangle = 0$, $\|e_J + e_R\| = \varepsilon = \sigma\|e_J - \gamma\bar{z}\|$, and $\langle \gamma\bar{z}, e_R - e_J\rangle = r\|e_R - e_J\|$, so every inequality in the proof holds with equality and $\|\gamma\bar{z} + e_J + 2e_R\|^2 = \Phi(\varepsilon) = G_\sigma^2 r^2$. Such a configuration is realized by an accepted pair: for any $\bar{y}$, the two-point set $\{(\bar{y}, \bar{z}), (\bar{y} + e_J, \bar{z} + e_R/\gamma)\}$ is monotone and therefore extends to a maximal monotone operator B [Bauschke and Combettes, 2017, Theorem 20.21], for which $\bar{y} = J(w)$ and $\gamma\bar{z} = R(w)$ at $w = \bar{y} + \gamma\bar{z}$, and $(y, z) = (\bar{y} + e_J, \bar{z} + e_R/\gamma)$ is a graph pair at w with residual $e = e_J + e_R$ that passes the relative test with equality. For $\sigma \le \sigma_1$ the extremal configuration is $e_J = 0$, $e_R = \sigma R(w)$: the inner solver returns the exact resolvent point $y = J(w)$, but with $\gamma z = (1 + \sigma)R(w)$. For $\sigma \ge \sigma_2$ it is $e_R = 0$, $e_J = -\frac{\sigma}{1-\sigma}R(w)$.

The constant H_σ in (10b) is sharp as well, and here we can construct an explicit operator. Let $\mathcal{H} = \mathcal{D} = \mathbb{R}^2$, $C = I$, and $A = B = \partial f$ with $f(y) = \frac{r}{\gamma}(|y_1| + \sigma H_\sigma |y_2|)$, a weighted ℓ_1 norm, so that J is the soft-thresholding map $J(w) = (\mathbf{sign}(w_1)\max\{|w_1| - r, 0\}, \mathbf{sign}(w_2)\max\{|w_2| - \sigma H_\sigma r, 0\})$. Fix $w_k = (w_1, 0)$ with $w_1 \ge \frac{2r}{1-\sigma^2}$, so that $J(w_k) = (w_1 - r, 0)$ and $R(w_k) = (r, 0)$, and let the accepted pair be

$$y_k = J(w_k) - \frac{\sigma^2}{1-\sigma^2}R(w_k) = \left(w_1 - \frac{r}{1-\sigma^2}, 0\right), \qquad \gamma z_k = (r, \sigma H_\sigma r),$$

a graph pair because $\gamma B(y_k) = \{r\} \times [-\sigma H_\sigma r, \sigma H_\sigma r]$ (as $y_{k,1} > 0$ and $y_{k,2} = 0$). In the notation of the proof, $e_J = -\frac{\sigma^2}{1-\sigma^2}R(w_k)$ and $e_R = (0, \sigma H_\sigma r) \perp e_J$, so $\|e_R\| = \frac{\sigma r}{\sqrt{1-\sigma^2}}$ attains the bound on $\|e_R\|$, and the residual $e_k = e_J + e_R$ satisfies $\|e_k\|^2 = \frac{\sigma^4 r^2}{(1-\sigma^2)^2} + \frac{\sigma^2 r^2}{1-\sigma^2} = \frac{\sigma^2 r^2}{(1-\sigma^2)^2} = \sigma^2\|y_k - w_k\|^2$: the relative test passes with equality (and the terminal test fails whenever $\gamma\delta < \|\gamma z_k\| = \frac{r}{\sqrt{1-\sigma^2}}$).

Then $w_{k+1} = w_k - \gamma z_k = (w_1 - r, -\sigma H_\sigma r)$ with $w_1 - r > r$, so $J(w_{k+1}) = (w_1 - 2r, 0)$, $R(w_{k+1}) = (r, -\sigma H_\sigma r) = R(w_k) - e_R$, and

$$\|R(w_{k+1})\|^2 = r^2 + \sigma^2 H_\sigma^2 r^2 = \|R(w_k)\|^2 + \|e_R\|^2 = \frac{r^2}{1-\sigma^2} = H_\sigma^2 \|R(w_k)\|^2,$$

so both inequalities leading to (38) hold with equality. The extremal configurations for the two constants differ: for G_σ (with $\sigma \le \sigma_1$) the error e_R in γz_k is aligned with $R(w_k)$, whereas for H_σ it is orthogonal to $R(w_k)$ and as large as the relative test allows, since (38) adds it in quadrature. The first coordinate of this example is the instance of Remark B.1. □

Remark B.1 (Sharpness of Theorems 4.1, 4.2 and 4.4). The per-step bound (17) is attained, for every $\sigma \in (0,1)$, s, α, and C_0, by the following instance. Let $\mathcal{H} = \mathcal{D} = \mathbb{R}$, $C = I$, $\gamma = 1$, and $A = B = \partial(r|\cdot|)$ with $r > \delta$, so that $J(w) = w - r$ and $R(w) = r$ for $w > r$, and let $w_0 > r$. At an outer iteration with $w_k > r$, so that $\|R(w_k)\| = r$, let the inner procedure return

$$\begin{aligned} y^l &= J(w_k) + t_l R(w_k), \\ z^l &= R(w_k)/\gamma, \\ t_l &:= \sqrt{C_0}\,\alpha^{l/2} t_0 \quad (l \ge 1), \\ t_0 &:= \frac{\|\gamma z^0 + y^0 - w_k\|}{r}, \end{aligned}$$

that is, y^l moves along the segment from $J(w_k)$ toward w_k while the operator value stays exact. Then $y^l > 0$, so $(y^l, z^l) \in \mathbf{gra}\, B$, and the residual is $e^l = t_l R(w_k)$. Thus, $\|e^l\|^2 = C_0 \alpha^l \|e^0\|^2$ for $l \ge 1$. The relative test $\|e^l\| \le \sigma \|y^l - w_k\| = \sigma |1 - t_l| r$ holds if and only if $t_l \le \frac{\sigma}{1+\sigma}$; i.e., if and only if $\|e^l\| \le \frac{\sigma}{1+\sigma}\|R(w_k)\|$; and the terminal test fails because $\|z^l\| = r > \delta$. Hence the inner loop exits at the first check at which the sufficient condition used in the proof of Theorem 4.1 holds, and $N_k = s \left\lceil \frac{(\log \tilde{\beta}_k)_+}{s \log(1/\alpha)} \right\rceil$ exactly (whenever $\|e^0\| > \frac{\sigma}{1+\sigma}\|R(w_k)\|$, which is the case for $k \ge 1$ below). Since z is exact, the outer step is the exact proximal step $w_{k+1} = J(w_k)$, and the next warm start is $e^0_{k+1} = e_k + \gamma z_k = (1 + t_{\text{exit}}) R(w_k)$ with $t_{\text{exit}} \le \frac{\sigma}{1+\sigma}$, so that $\|e^0_{k+1}\| \le \frac{1+2\sigma}{1+\sigma}\|R(w_k)\|$ and, for $k \ge 1$,

$$\tilde{\beta}_k \le \frac{(1+2\sigma)^2 C_0}{\sigma^2} \frac{\|R(w_{k-1})\|^2}{\|R(w_k)\|^2} = \frac{D}{(1+\sigma)^2} \frac{\|R(w_{k-1})\|^2}{\|R(w_k)\|^2} \qquad (\sigma \le \sigma_1),$$

with equality in the inequality when $t_{\text{exit}} = \frac{\sigma}{1+\sigma}$, which is approached for $s = 1$ as $\alpha \to 1$. A randomized variant of the same instance, in which the residual decreases deterministically to just above the threshold $\frac{\sigma}{1+\sigma}\|R(w_k)\|$ and then, at each inner iteration, is either solved exactly with probability $1 - \alpha$ or left unchanged, makes Markov's inequality in the proof of Theorem 4.1 tight at every check. It shows that the bound on $\mathbf{E}_k[N_k]$ there, and with it the per-iteration constant $\frac{s}{1-\alpha^s}$ of Theorem 4.2, can be approached arbitrarily closely.

C Proof for Accelerated Inner Solves

Proof of Proposition 5.2. Fix an outer iteration k, and let $x_k^\star$ be the unique minimizer of φ_k. As in the gradient descent case, φ_k is μ-strongly convex and L_φ-smooth, with $\mu = \frac{1}{\gamma}$ and $\frac{L_\varphi}{\mu} = \kappa$. The graph pair $(x^l, \nabla f(x^l))$ has residual $e^l = \gamma \nabla \varphi_k(x^l)$. The standard estimate for Nesterov's method with zero initial momentum gives

$$\varphi_k(x^l) - \varphi_k(x_k^\star) \le \left(1 - \frac{1}{\sqrt{\kappa}}\right)^l \left(\varphi_k(x^0) - \varphi_k(x_k^\star) + \frac{\mu}{2}\|x^0 - x_k^\star\|^2\right). \tag{39}$$

This estimate follows from the estimating sequence analysis of Nesterov [2013, Theorem 2.2.3], with initial curvature μ and constant acceleration parameter $\sqrt{\frac{\mu}{L_\varphi}}$. Strong convexity yields

$$\frac{\mu}{2}\|x^0 - x_k^\star\|^2 \le \varphi_k(x^0) - \varphi_k(x_k^\star) \le \frac{1}{2\mu}\|\nabla\varphi_k(x^0)\|^2.$$

Smoothness gives $\|\nabla\varphi_k(x^l)\|^2 \le 2L_\varphi(\varphi_k(x^l) - \varphi_k(x_k^\star))$. Combining these inequalities with (39), we obtain

$$\begin{aligned}\|e^l\|^2 &\le 2\gamma^2 L_\varphi \left(\varphi_k(x^l) - \varphi_k(x_k^\star)\right) \\ &\le \frac{2L_\varphi}{\mu}\left(1 - \frac{1}{\sqrt{\kappa}}\right)^l \gamma^2\|\nabla\varphi_k(x^0)\|^2 \\ &= 2\kappa\left(1 - \frac{1}{\sqrt{\kappa}}\right)^l \|e^0\|^2.\end{aligned}$$

This deterministic residual bound holds for every inner iteration and every outer warm start, proving (21).

Substituting these constants into Theorems 4.2 and 4.4 gives the stated coefficient of δ^{-2}. The parameter-selection argument in the proof of Proposition 5.1 uses only that $C_0 \ge 1$ and that α and C_0 do not depend on σ. These properties hold here, so that argument gives uniqueness, the equation (20), and $\sigma^\star < (3/2 + \chi + \log(2\kappa))^{-1}$. The final inequality follows from $\chi \ge s\log(1/\alpha)$. The remaining terms have the same dependence on δ as in the gradient descent case. □

D Additional Numerical Results

This appendix collects figures that supplement Section 6 without being needed to follow it: magnified views of the convergence figures and the convergence of the rotation-operator sweep. Its subsections mirror those of Section 6, so that further details of each study can be added in place.

D.1 Inexact Proximal Point

Figure 11 magnifies the outer-iteration panels of Figure 2 on the early iterations and on the final approach, where the curves of the σ sweep coincide at full scale.

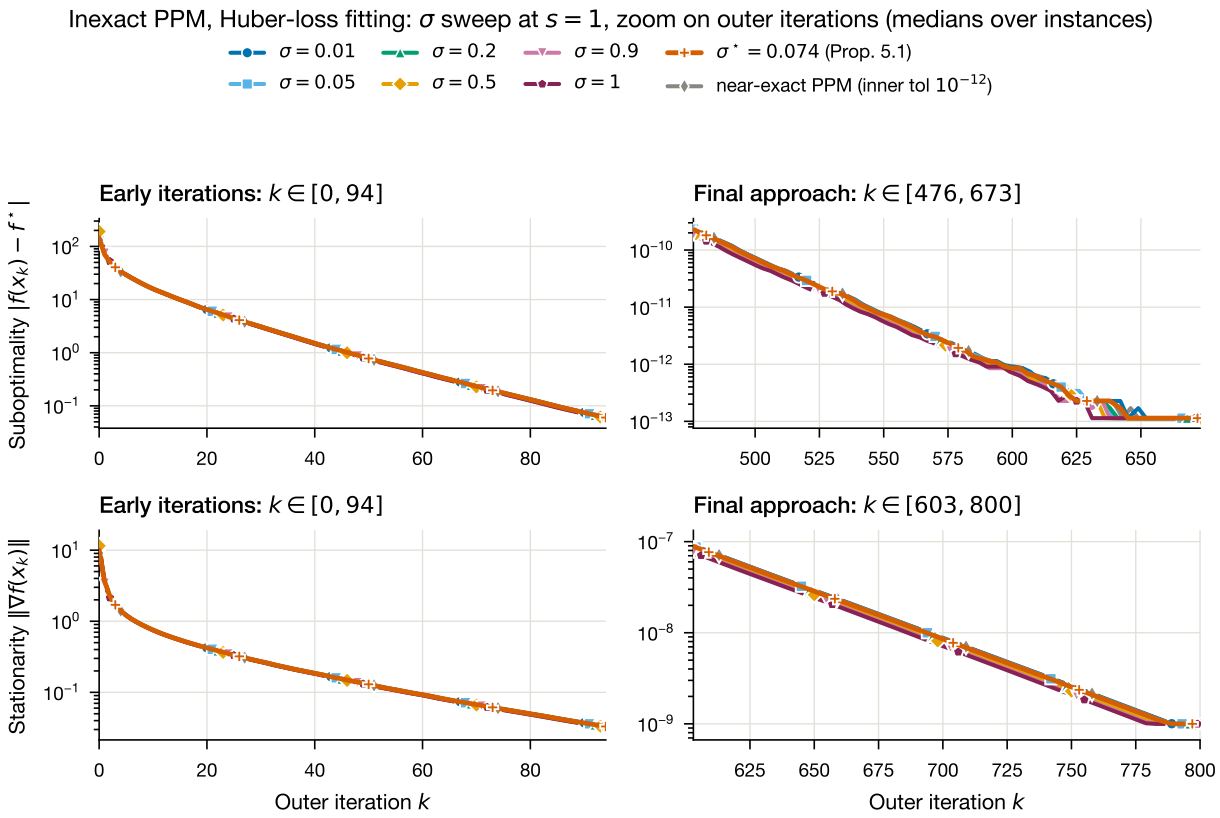


Figure 11: The outer-iteration panels of Figure 2 on two windows: the first 12% of the median run (left) and the final approach of each metric to its plateau (right), as medians over the 100 instances only, so that curves that coincide at full scale can be told apart.

D.2 Inexact ADMM

Figure 12 does the same for Figure 6.

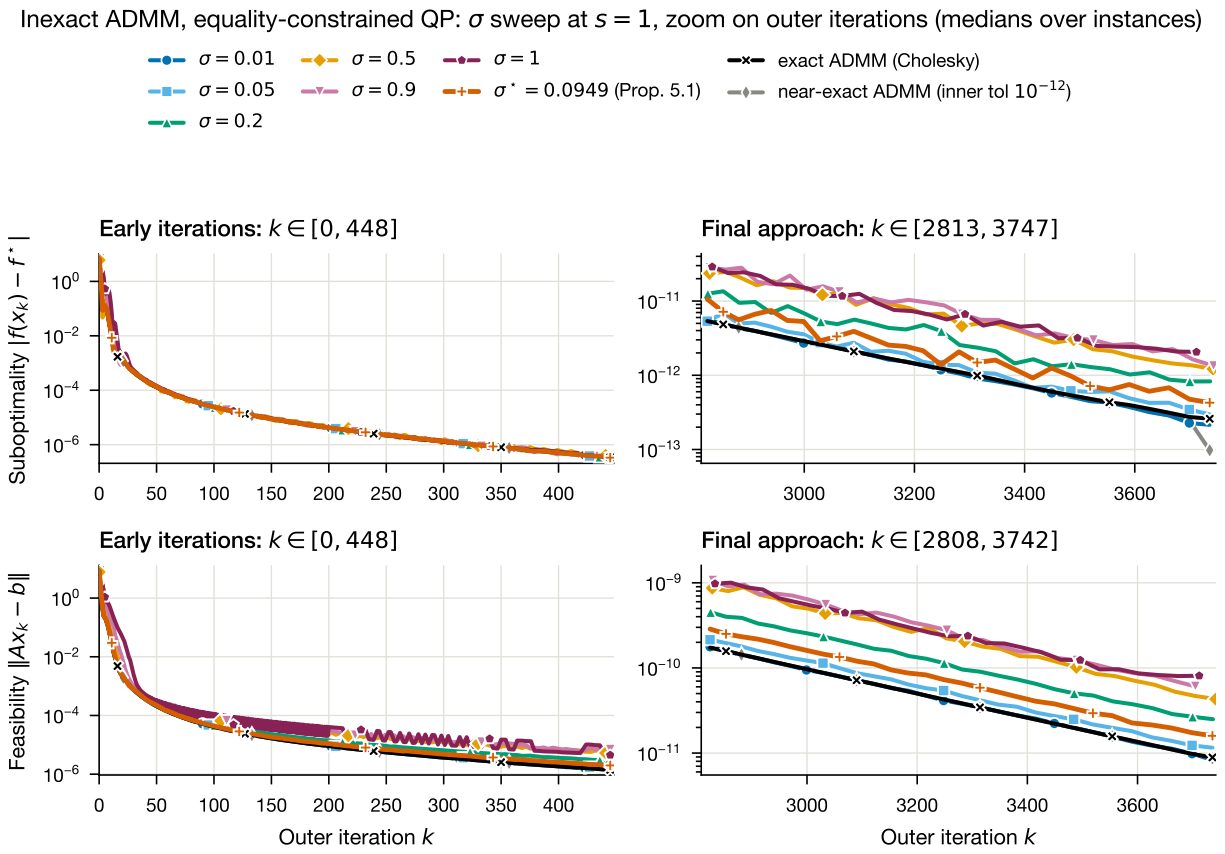


Figure 12: The outer-iteration panels of Figure 6 on two windows: the first 12% of the median run (left) and the final approach of each metric to its plateau (right), as medians over the 100 instances only, so that curves that coincide at full scale can be told apart.

D.3 Rotation Operator

Figure 13 shows the convergence of the σ sweep on the rotation operator of Figure 9 in the layout of the main studies.

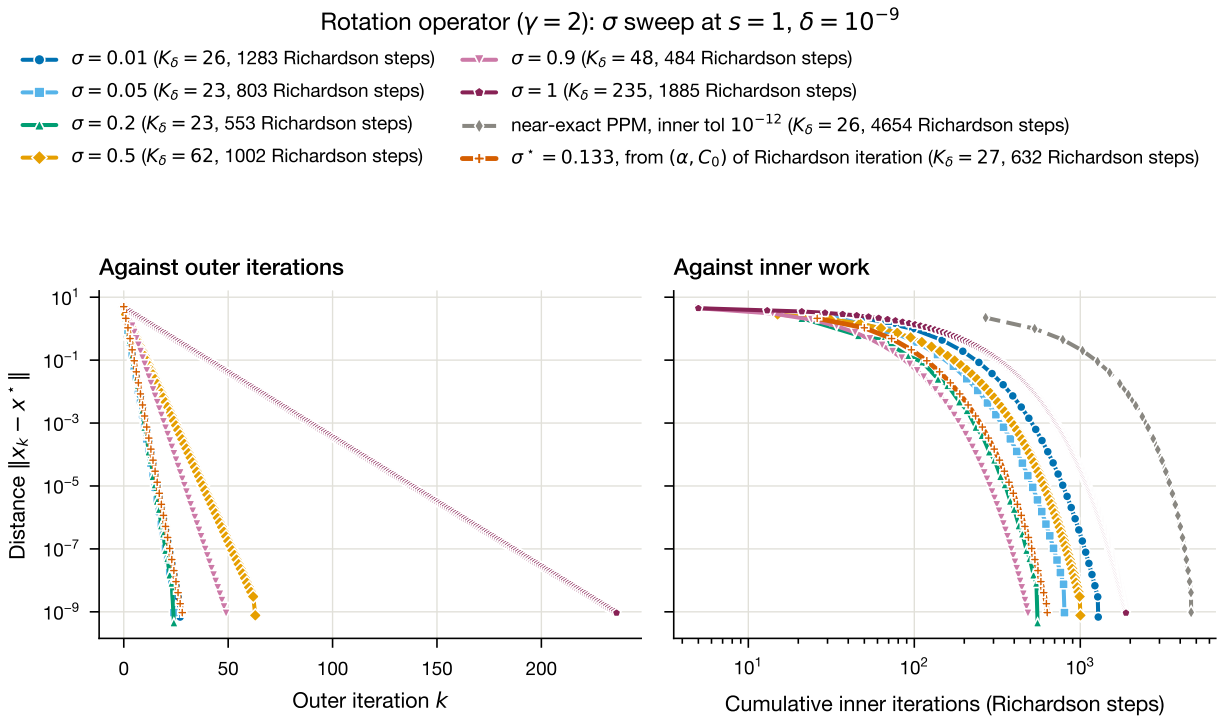


Figure 13: Convergence on the rotation operator of Figure 9 for the σ sweep at $s = 1$, for $\sigma^\star$, and for the near-exact method, in the layout of Figure 2: the distance $\|x_k - x^\star\| = \|B(x_k)\|$ against outer iterations (left) and against cumulative Richardson steps (right). There is a single instance, so each curve is one run and there are no bands. The near-exact method stops after 26 outer iterations and the $\sigma = 1$ run after 235.

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